PHARMACY MANAGEMENT SYSTEM

Object-Oriented Programming - Mini Project Report

Submitted in partial fulfillment of the requirements for the
Bachelor of Technology in Computer Science and Engineering

Project Title	Pharmacy Management System
Subject	Object-Oriented Programming (OOP)
Academic Year	2025 - 2026

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Introduction

When we started looking for a project idea, one of our team members mentioned his uncle owns a small pharmacy near his house. The pharmacy still runs on paper - a thick register for stock entries, handwritten bills, prescriptions tucked into folders. He said the staff spends at least an hour every day just searching through records. That gave us the idea. A pharmacy is actually a perfect use case for everything we were learning in our OOP lab - it has multiple types of data, multiple users with different access levels, operations that need to happen in the background, and files that need to be saved and loaded between sessions.

So we built a desktop application for pharmacy management using Java and JavaFX. The system handles six main areas: medicine inventory, billing and invoices, customer records, supplier information, prescriptions, and a reports section. Staff can search for any medicine instantly, build a bill by adding items one by one with automatic price calculation, and save prescription records linked to specific patients. Two monitoring threads run in the background the entire time the application is open - one checks for medicines running low on stock, the other watches expiry dates - so the staff get warnings without having to check anything manually.

From a learning perspective, this project made OOP concepts feel real rather than textbook exercises. We needed encapsulation to protect medicine and billing data from being changed accidentally. We needed inheritance when building threads and the main application class. We needed method overloading in the billing logic where discount can be applied as either a percentage or a flat amount. Interfaces gave us a clean way to pass results back from background threads to the UI. Generics let us write a single file-storage class that works for medicines, customers, suppliers, bills, and prescriptions without duplicating any code. Every concept earned its place.

Problem Statement

Pharmacy Management System with Inventory Control and Billing Automation

The core problem with paper-based pharmacy management is not just that it is slow. It is that mistakes are invisible until they cause damage. A billing error in a handwritten invoice is discovered only when the patient questions it. An expired medicine batch is noticed only when someone picks it up to dispense it. A stock shortage is caught only when the shelf is empty and a customer is waiting. None of these are acceptable situations in a healthcare setting, yet they are a daily reality for pharmacies that have not upgraded their systems.

Our system targets these three failure points directly. Every invoice is calculated by the software, so arithmetic errors are not possible. Expiry dates are monitored continuously by a background thread that flags anything expiring within thirty days, giving staff time to act before the medicine becomes unusable. Stock levels trigger a separate alert the moment any item drops below ten units. Beyond these alerts, the system also stores prescription history per patient, tracks which supplier provided which batch, and generates text-format invoice files that can be printed or archived. The whole application is designed to run on a standard pharmacy counter PC with no internet connection required.

Implementation Details

The entire codebase is in Java, with JavaFX handling the visual interface. We picked JavaFX because it gives proper desktop UI components out of the box - tables that sort when you click a column header, form layouts that align fields neatly, styled buttons, dropdown selectors - all without needing a browser or a web framework. The 32 source files are split into four packages based on

what they do rather than how they look. Model classes are just data containers. Service classes hold all the logic and have no knowledge of the interface. UI classes build and display screens. Utility classes provide shared tools like ID generation and file storage that every other package depends on.

We initially wrote the inventory screen so that it handled its own data loading and saving. It quickly became a 600-line class that was difficult to follow. Splitting the data operations out into a separate service class cut the screen file in half and made both halves individually readable. That experience shaped how we structured the rest of the project. Every screen does only three things: build the layout, react to user input, and call the right service method. The service layer does the actual work. Since background threads are also reading and writing the same data that the service classes manage, every write operation inside those classes is locked - only one thread can execute it at a time, which prevents data from getting corrupted when two operations overlap.

All 32 source files in this project:

- Medicine.java
- MedicineCategory.java
- UserRole.java
- Customer.java
- Supplier.java
- Bill.java
- BillItem.java
- Prescription.java
- PrescriptionItem.java
- User.java
- InventoryManager.java
- BillingService.java
- CustomerService.java
- SupplierService.java
- PrescriptionService.java
- UserService.java
- ExpiryAlertThread.java
- StockAlertThread.java
- BillingThread.java
- BackupService.java
- PharmacyApp.java
- LoginScreen.java
- DashboardScreen.java
- InventoryScreen.java
- BillingScreen.java
- SupplierScreen.java
- CustomerScreen.java
- PrescriptionScreen.java
- ReportsScreen.java
- DataStore.java
- IDGenerator.java
- SampleDataLoader.java

Detailed Explanation of Each File

Model Package (com.pharmacy.model)

1. Medicine.java

Medicine.java is the data class for a single stock item. Each medicine object stores its ID, name, manufacturer, category, batch number, expiry date, quantity, price, and supplier ID. We deliberately declared quantity as Integer (capital I) and price as Double (capital D) rather than using the primitive forms int and double. The reason was to have autoboxing and unboxing visible in the code - whenever we assign a plain number to those fields, Java silently wraps it into an object, and when we use it in a calculation, it unwraps it back. The class has two constructors: one that takes all nine fields including the supplier ID, and another shorter one that leaves the supplier ID out for cases where it is not known yet. That is constructor overloading. There is also a compareTo method which makes it possible to sort a list of medicines alphabetically by name without writing any extra comparison logic at the call site.

Code for Medicine.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Medicine model class representing a medicine in the pharmacy.
* Demonstrates: Encapsulation, Serialization, Constructors, Wrapper classes.
*/
public class Medicine implements Serializable, Comparable<Medicine> {
private static final long serialVersionUID = 1L;
private String medicineId;
private String name;
private String manufacturer;
private MedicineCategory category;
private String batchNumber;
private String expiryDate; // Format: YYYY-MM-DD
private Integer quantity; // Wrapper class demonstration
private Double price; // Wrapper class demonstration
private String supplierId;
// Default constructor
public Medicine() {
}
// Parameterized constructor
public Medicine(String medicineId, String name, String manufacturer,
MedicineCategory category, String batchNumber,
String expiryDate, int quantity, double price, String supplierId) {
this.medicineId = medicineId;
this.name = name;
this.manufacturer = manufacturer;
this.category = category;
this.batchNumber = batchNumber;
this.expiryDate = expiryDate;
this.quantity = quantity; // Autoboxing: int -> Integer
this.price = price; // Autoboxing: double -> Double
this.supplierId = supplierId;
}
// Constructor overloading (without supplier)
```

```
public Medicine(String medicineId, String name, String manufacturer,
MedicineCategory category, String batchNumber,
String expiryDate, int quantity, double price) {
this(medicineId, name, manufacturer, category, batchNumber,
expiryDate, quantity, price, "");
}
// Getters and Setters (Encapsulation)
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getName() { return name; }
public void setName(String name) { this.name = name; }
public String getManufacturer() { return manufacturer; }
public void setManufacturer(String manufacturer) { this.manufacturer = manufacturer; }
public MedicineCategory getCategory() { return category; }
public void setCategory(MedicineCategory category) { this.category = category; }
public String getBatchNumber() { return batchNumber; }
public void setBatchNumber(String batchNumber) { this.batchNumber = batchNumber; }
public String getExpiryDate() { return expiryDate; }
public void setExpiryDate(String expiryDate) { this.expiryDate = expiryDate; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public Double getPrice() { return price; }
public void setPrice(Double price) { this.price = price; }
public String getSupplierId() { return supplierId; }
public void setSupplierId(String supplierId) { this.supplierId = supplierId; }
// Unboxing demonstration
public int getQuantityPrimitive() {
return quantity; // Unboxing: Integer -> int
}
public double getPricePrimitive() {
return price; // Unboxing: Double -> double
}
// Comparable implementation for sorting
@Override
public int compareTo(Medicine other) {
return this.name.compareToIgnoreCase(other.name);
}
@Override
public String toString() {
return medicineId + " | " + name + " | " + category + " | Qty: " + quantity
+ " | Rs." + String.format("%.2f", price) + " | Exp: " + expiryDate;
}
}
```

2. MedicineCategory.java

This is an enum with eight values: TABLET, CAPSULE, SYRUP, INJECTION, OINTMENT, DROPS, INHALER, and POWDER. Before we introduced this, category was stored as a plain String, which caused problems - one part of the code would write "Tablet" and another would write "tablet" and they would not match. An enum fixes that completely because there is only one possible value and Java enforces it at compile time. Each enum value also carries a human-readable display name so the UI can show "Tablet" instead of "TABLET" without any extra formatting code.

Code for MedicineCategory.java:

```
package com.pharmacy.model;
/**
* Enum representing different categories of medicines.
* Demonstrates: Enum usage as required by OSDL syllabus.
*/
public enum MedicineCategory {
TABLET("Tablet"),
CAPSULE("Capsule"),
SYRUP("Syrup"),
INJECTION("Injection"),
OINTMENT("Ointment"),
DROPS("Drops"),
INHALER("Inhaler"),
POWDER("Powder");
private final String displayName;
// Enum constructor
MedicineCategory(String displayName) {
this.displayName = displayName;
}
public String getDisplayName() {
return displayName;
}
@Override
public String toString() {
return displayName;
}
}
```

3. UserRole.java

The system supports two account types: ADMIN for full access and PHARMACIST for a restricted view. Both are defined in this enum. Each value carries a boolean field that says whether that role is allowed to manage user accounts. Using an enum here instead of a String or integer constant means the compiler will catch any typo in a role check at build time rather than at runtime.

Code for UserRole.java:

```
package com.pharmacy.model;
/**
* Enum for role-based access control.
* Demonstrates: Enum with methods.
*/
public enum UserRole {
ADMIN("Administrator", true),
PHARMACIST("Pharmacist", false);
private final String displayName;
private final boolean canManageUsers;
UserRole(String displayName, boolean canManageUsers) {
this.displayName = displayName;
this.canManageUsers = canManageUsers;
}
public String getDisplayName() {
return displayName;
}
public boolean canManageUsers() {
return canManageUsers;
```

```
}
}
```

4. Customer.java

Stores five details for each patient: a system-generated ID, their name, phone number, email, and address. The class is marked as serialisable, which is the flag Java needs to be able to write the object to a file and read it back correctly on the next startup. A toString method returns a compact one-line summary that is used in tables and status messages.

Code for Customer.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Customer/Patient model class.
* Demonstrates: Encapsulation, Serialization.
*/
public class Customer implements Serializable {
private static final long serialVersionUID = 1L;
private String customerId;
private String name;
private String phone;
private String email;
private String address;
// Default constructor
public Customer() {
}
// Parameterized constructor
public Customer(String customerId, String name, String phone, String email, String address) {
this.customerId = customerId;
this.name = name;
this.phone = phone;
this.email = email;
this.address = address;
}
// Getters and Setters
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getName() { return name; }
public void setName(String name) { this.name = name; }
public String getPhone() { return phone; }
public void setPhone(String phone) { this.phone = phone; }
public String getEmail() { return email; }
public void setEmail(String email) { this.email = email; }
public String getAddress() { return address; }
public void setAddress(String address) { this.address = address; }
@Override
public String toString() {
return customerId + " | " + name + " | " + phone;
}
}
```

5. Supplier.java

A supplier entry covers the company name, a specific contact person at that company, their phone and email, and the company address. Medicines in the inventory link back to a supplier through a supplier ID stored in the medicine record itself, so you can trace any batch back to where it came from. Serialisable like all the other model classes.

Code for Supplier.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Supplier model class.
* Demonstrates: Encapsulation, Serialization.
*/
public class Supplier implements Serializable {
private static final long serialVersionUID = 1L;
private String supplierId;
private String companyName;
private String contactPerson;
private String phone;
private String email;
private String address;
// Default constructor
public Supplier() {
}
// Parameterized constructor
public Supplier(String supplierId, String companyName, String contactPerson,
String phone, String email, String address) {
this.supplierId = supplierId;
this.companyName = companyName;
this.contactPerson = contactPerson;
this.phone = phone;
this.email = email;
this.address = address;
}
// Getters and Setters
public String getSupplierId() { return supplierId; }
public void setSupplierId(String supplierId) { this.supplierId = supplierId; }
public String getCompanyName() { return companyName; }
public void setCompanyName(String companyName) { this.companyName = companyName; }
public String getContactPerson() { return contactPerson; }
public void setContactPerson(String contactPerson) { this.contactPerson = contactPerson; }
public String getPhone() { return phone; }
public void setPhone(String phone) { this.phone = phone; }
public String getEmail() { return email; }
public void setEmail(String email) { this.email = email; }
public String getAddress() { return address; }
public void setAddress(String address) { this.address = address; }
@Override
public String toString() {
return supplierId + " | " + companyName + " | " + contactPerson;
}
}
```

6. Bill.java

A complete billing record contains the customer details, the transaction date, the username of whoever processed the bill, and a list of all the line items. The discount feature is written using method overloading - there are two methods both named applyDiscount, but they take different parameters. One takes just a percentage value. The other takes an amount and a boolean flag: if the flag is true, the amount is treated as a fixed rupee deduction; if false, it is treated as a percentage. This is one of the clearer demonstrations of polymorphism in the project.

Code for Bill.java:

```
package com.pharmacy.model;
import java.io.Serializable;
import java.util.ArrayList;
/**
* Bill/Invoice model class for sales transactions.
* Demonstrates: Encapsulation, Serialization, ArrayList, Wrapper classes.
*/
public class Bill implements Serializable {
private static final long serialVersionUID = 1L;
private String billId;
private String customerId;
private String customerName;
private String date;
private ArrayList<BillItem> items;
private Double totalAmount; // Wrapper class
private Double discount;
private Double netAmount;
private String billedBy; // username of the person who billed
// Default constructor
public Bill() {
this.items = new ArrayList<>();
this.totalAmount = 0.0;
this.discount = 0.0;
this.netAmount = 0.0;
}
// Parameterized constructor
public Bill(String billId, String customerId, String customerName,
String date, String billedBy) {
this.billId = billId;
this.customerId = customerId;
this.customerName = customerName;
this.date = date;
this.billedBy = billedBy;
this.items = new ArrayList<>();
this.totalAmount = 0.0;
this.discount = 0.0;
this.netAmount = 0.0;
}
// Add item and recalculate total
public void addItem(BillItem item) {
items.add(item);
recalculate();
}
// Recalculate totals
public void recalculate() {
double total = 0.0;
for (BillItem item : items) {
```

```
total += item.getSubtotal(); // Unboxing
}
this.totalAmount = total; // Autoboxing
this.netAmount = totalAmount - discount;
}
// Method overloading: apply discount by percentage
public void applyDiscount(double percentage) {
this.discount = totalAmount * (percentage / 100.0);
this.netAmount = totalAmount - discount;
}
// Method overloading: apply discount by fixed amount with a flag
public void applyDiscount(double amount, boolean isFixed) {
if (isFixed) {
this.discount = amount;
} else {
this.discount = totalAmount * (amount / 100.0);
}
this.netAmount = totalAmount - discount;
}
// Getters and Setters
public String getBillId() { return billId; }
public void setBillId(String billId) { this.billId = billId; }
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getCustomerName() { return customerName; }
public void setCustomerName(String customerName) { this.customerName = customerName; }
public String getDate() { return date; }
public void setDate(String date) { this.date = date; }
public ArrayList<BillItem> getItems() { return items; }
public void setItems(ArrayList<BillItem> items) { this.items = items; }
public Double getTotalAmount() { return totalAmount; }
public void setTotalAmount(Double totalAmount) { this.totalAmount = totalAmount; }
public Double getDiscount() { return discount; }
public void setDiscount(Double discount) { this.discount = discount; }
public Double getNetAmount() { return netAmount; }
public void setNetAmount(Double netAmount) { this.netAmount = netAmount; }
public String getBilledBy() { return billedBy; }
public void setBilledBy(String billedBy) { this.billedBy = billedBy; }
@Override
public String toString() {
return billId + " | " + customerName + " | Rs." + String.format("%.2f", netAmount)
+ " | " + date;
}
}
```

7. BillItem.java

One line in a bill - one medicine, its quantity, unit price, and calculated subtotal. The quantity is stored as Integer and the prices as Double, both capital-letter object types rather than lowercase primitives. The constructor takes plain int and double values as parameters and assigns them to these fields, which is where autoboxing happens: Java wraps the primitive into the object form automatically without us having to call any conversion method.

Code for BillItem.java:

```
package com.pharmacy.model;
```

```
import java.io.Serializable;
/**
* Individual item in a bill.
* Demonstrates: Encapsulation, Serialization, Wrapper classes.
*/
public class BillItem implements Serializable {
private static final long serialVersionUID = 1L;
private String medicineId;
private String medicineName;
private Integer quantity; // Wrapper class
private Double unitPrice; // Wrapper class
private Double subtotal; // Wrapper class
public BillItem() {
}
public BillItem(String medicineId, String medicineName, int quantity, double unitPrice) {
this.medicineId = medicineId;
this.medicineName = medicineName;
this.quantity = quantity; // Autoboxing
this.unitPrice = unitPrice; // Autoboxing
this.subtotal = quantity * unitPrice; // Autoboxing result
}
// Getters and Setters
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getMedicineName() { return medicineName; }
public void setMedicineName(String medicineName) { this.medicineName = medicineName; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public Double getUnitPrice() { return unitPrice; }
public void setUnitPrice(Double unitPrice) { this.unitPrice = unitPrice; }
public Double getSubtotal() { return subtotal; }
public void setSubtotal(Double subtotal) { this.subtotal = subtotal; }
@Override
public String toString() {
return medicineName + " x" + quantity + " @ Rs." + String.format("%.2f", unitPrice)
+ " = Rs." + String.format("%.2f", subtotal);
}
}
```

8. Prescription.java

A prescription belongs to a specific patient and was written by a specific doctor on a specific date. This class holds all of that, plus any notes (dosage instructions, special warnings), and a list of PrescriptionItem objects - one for each medicine the doctor prescribed. The addItem method grows that list one entry at a time as items are added in the prescription screen.

Code for Prescription.java:

```
package com.pharmacy.model;
import java.io.Serializable;
import java.util.ArrayList;
/**
* Prescription model class linking customer to prescribed medicines.
* Demonstrates: Encapsulation, Serialization, ArrayList usage.
*/
public class Prescription implements Serializable {
```

```
private static final long serialVersionUID = 1L;
private String prescriptionId;
private String customerId;
private String customerName;
private String doctorName;
private String date;
private ArrayList<PrescriptionItem> items;
private String notes;
// Default constructor
public Prescription() {
this.items = new ArrayList<>();
}
// Parameterized constructor
public Prescription(String prescriptionId, String customerId, String customerName,
String doctorName, String date, String notes) {
this.prescriptionId = prescriptionId;
this.customerId = customerId;
this.customerName = customerName;
this.doctorName = doctorName;
this.date = date;
this.notes = notes;
this.items = new ArrayList<>();
}
// Add item to prescription
public void addItem(PrescriptionItem item) {
items.add(item);
}
// Getters and Setters
public String getPrescriptionId() { return prescriptionId; }
public void setPrescriptionId(String prescriptionId) { this.prescriptionId = prescriptionId; }
public String getCustomerId() { return customerId; }
public void setCustomerId(String customerId) { this.customerId = customerId; }
public String getCustomerName() { return customerName; }
public void setCustomerName(String customerName) { this.customerName = customerName; }
public String getDoctorName() { return doctorName; }
public void setDoctorName(String doctorName) { this.doctorName = doctorName; }
public String getDate() { return date; }
public void setDate(String date) { this.date = date; }
public ArrayList<PrescriptionItem> getItems() { return items; }
public void setItems(ArrayList<PrescriptionItem> items) { this.items = items; }
public String getNotes() { return notes; }
public void setNotes(String notes) { this.notes = notes; }
@Override
public String toString() {
return prescriptionId + " | Patient: " + customerName + " | Dr. " + doctorName + " | " + date;
}
}
```

9. PrescriptionItem.java

Each medicine in a prescription is its own PrescriptionItem. It records the medicine name, how many units were prescribed, and a dosage instruction string like "twice daily after meals". The dosage text is entered by the staff member at the time of creating the prescription - it is not pulled from the inventory because dosage instructions vary by patient even for the same medicine.

Code for PrescriptionItem.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* Individual item within a prescription.
* Demonstrates: Encapsulation, Serialization.
*/
public class PrescriptionItem implements Serializable {
private static final long serialVersionUID = 1L;
private String medicineId;
private String medicineName;
private Integer quantity;
private String dosage;
public PrescriptionItem() {
}
public PrescriptionItem(String medicineId, String medicineName, int quantity, String dosage) {
this.medicineId = medicineId;
this.medicineName = medicineName;
this.quantity = quantity; // Autoboxing
this.dosage = dosage;
}
public String getMedicineId() { return medicineId; }
public void setMedicineId(String medicineId) { this.medicineId = medicineId; }
public String getMedicineName() { return medicineName; }
public void setMedicineName(String medicineName) { this.medicineName = medicineName; }
public Integer getQuantity() { return quantity; }
public void setQuantity(Integer quantity) { this.quantity = quantity; }
public String getDosage() { return dosage; }
public void setDosage(String dosage) { this.dosage = dosage; }
@Override
public String toString() {
return medicineName + " x" + quantity + " (" + dosage + ")";
}
}
```

10. User.java

Login accounts for pharmacy staff. Each User object stores a username, a password, the person's full name, and their role as a UserRole enum value. The authenticate method inside the class accepts a password string and returns true or false depending on whether it matches. We put that check inside the model class deliberately so the login screen does not need to know anything about how passwords are compared - it just calls the method and acts on the result.

Code for User.java:

```
package com.pharmacy.model;
import java.io.Serializable;
/**
* User model for login and role-based access.
* Demonstrates: Encapsulation, Enum usage, Serialization.
*/
public class User implements Serializable {
private static final long serialVersionUID = 1L;
private String username;
private String password;
```

```
private String fullName;
private UserRole role;
// Default constructor
public User() {
}
// Parameterized constructor
public User(String username, String password, String fullName, UserRole role) {
this.username = username;
this.password = password;
this.fullName = fullName;
this.role = role;
}
// Validate login credentials
public boolean authenticate(String inputPassword) {
return this.password.equals(inputPassword);
}
// Getters and Setters
public String getUsername() { return username; }
public void setUsername(String username) { this.username = username; }
public String getPassword() { return password; }
public void setPassword(String password) { this.password = password; }
public String getFullName() { return fullName; }
public void setFullName(String fullName) { this.fullName = fullName; }
public UserRole getRole() { return role; }
public void setRole(UserRole role) { this.role = role; }
@Override
public String toString() {
return username + " (" + role.getDisplayName() + ")";
}
}
```

Service Package (com.pharmacy.service)

11. InventoryManager.java

This is the most heavily used service class in the whole project. It holds all medicines in two data structures at once. There is a list, which keeps medicines in the order they were added and can be iterated for display. There is also a map keyed by medicine ID, which gives you any specific medicine in constant time without scanning through the whole list. Both are kept in sync on every add, update, and delete. Deletion is done using Java's iterator cursor rather than removing inside a regular loop, because removing from a list while iterating it normally throws an error. Since the expiry alert thread and the stock alert thread both read inventory data while the user might be adding or editing, every write method is locked - only one caller can run it at a time. Sorting is available in three ways: by name using the compareTo method in Medicine.java, by price using a custom comparator, and by expiry date using another comparator.

Code for InventoryManager.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import com.pharmacy.model.MedicineCategory;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;
```

```
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages the medicine inventory.
* Demonstrates: ArrayList, HashMap, Iterator, Collections.sort, Generics, Synchronization.
*/
public class InventoryManager {
private ArrayList<Medicine> medicineList;
private HashMap<String, Medicine> medicineMap; // Maps medicineId -> Medicine object
private DataStore<Medicine> dataStore;
private static final String DATA_FILE = "data/medicines.dat";
private static final String BACKUP_FILE = "data/backup_medicines.txt";
private static final int LOW_STOCK_THRESHOLD = 10;
public InventoryManager() {
this.dataStore = new DataStore<>(DATA_FILE);
this.medicineList = dataStore.loadAll();
this.medicineMap = new HashMap<>();
// Populate HashMap from loaded list
for (Medicine med : medicineList) {
medicineMap.put(med.getMedicineId(), med);
}
}
// Add medicine (synchronized for thread safety)
public synchronized void addMedicine(Medicine medicine) {
medicineList.add(medicine);
medicineMap.put(medicine.getMedicineId(), medicine);
saveData();
}
// Update medicine
public synchronized void updateMedicine(Medicine medicine) {
medicineMap.put(medicine.getMedicineId(), medicine);
// Update in list as well
for (int i = 0; i < medicineList.size(); i++) {
if (medicineList.get(i).getMedicineId().equals(medicine.getMedicineId())) {
medicineList.set(i, medicine);
break;
}
}
saveData();
}
// Delete medicine using Iterator
public synchronized void deleteMedicine(String medicineId) {
Iterator<Medicine> iterator = medicineList.iterator();
while (iterator.hasNext()) {
Medicine med = iterator.next();
if (med.getMedicineId().equals(medicineId)) {
iterator.remove();
break;
}
}
medicineMap.remove(medicineId);
saveData();
}
// Get medicine by ID using HashMap
public Medicine getMedicineById(String medicineId) {
```

```
return medicineMap.get(medicineId);
}
// Search medicines by name
public ArrayList<Medicine> searchByName(String name) {
ArrayList<Medicine> results = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getName().toLowerCase().contains(name.toLowerCase())) {
results.add(med);
}
}
return results;
}
// Filter by category
public ArrayList<Medicine> filterByCategory(MedicineCategory category) {
ArrayList<Medicine> results = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getCategory() == category) {
results.add(med);
}
}
return results;
}
// Sort by name using Collections.sort (Comparable)
public ArrayList<Medicine> getSortedByName() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted); // Uses Medicine's compareTo
return sorted;
}
// Sort by price using Comparator
public ArrayList<Medicine> getSortedByPrice() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted, new Comparator<Medicine>() {
@Override
public int compare(Medicine m1, Medicine m2) {
return Double.compare(m1.getPrice(), m2.getPrice());
}
});
return sorted;
}
// Sort by expiry date
public ArrayList<Medicine> getSortedByExpiry() {
ArrayList<Medicine> sorted = new ArrayList<>(medicineList);
Collections.sort(sorted, new Comparator<Medicine>() {
@Override
public int compare(Medicine m1, Medicine m2) {
return m1.getExpiryDate().compareTo(m2.getExpiryDate());
}
});
return sorted;
}
// Get low stock medicines
public ArrayList<Medicine> getLowStockMedicines() {
ArrayList<Medicine> lowStock = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getQuantity() < LOW_STOCK_THRESHOLD) {
lowStock.add(med);
```

```
}
}
return lowStock;
}
// Get expired medicines (compare with current date string YYYY-MM-DD)
public ArrayList<Medicine> getExpiredMedicines(String currentDate) {
ArrayList<Medicine> expired = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getExpiryDate().compareTo(currentDate) <= 0) {
expired.add(med);
}
}
return expired;
}
// Get medicines expiring soon (within 30 days approximation)
public ArrayList<Medicine> getExpiringSoon(String currentDate, String thresholdDate) {
ArrayList<Medicine> expiring = new ArrayList<>();
for (Medicine med : medicineList) {
if (med.getExpiryDate().compareTo(currentDate) > 0
&& med.getExpiryDate().compareTo(thresholdDate) <= 0) {
expiring.add(med);
}
}
return expiring;
}
// Reduce stock (for billing)
public synchronized boolean reduceStock(String medicineId, int quantity) {
Medicine med = medicineMap.get(medicineId);
if (med != null && med.getQuantity() >= quantity) {
med.setQuantity(med.getQuantity() - quantity);
saveData();
return true;
}
return false;
}
// Get all medicines
public ArrayList<Medicine> getAllMedicines() {
return medicineList;
}
// Save data
private void saveData() {
dataStore.saveAll(medicineList);
}
// Create backup
public void createBackup() {
dataStore.backupToText(medicineList, BACKUP_FILE);
}
public int getMedicineCount() {
return medicineList.size();
}
}
```

12. BillingService.java

All billing operations go through here. When a bill is submitted, this class checks each item against current stock levels, rejects the whole transaction if anything is short, otherwise deducts all

quantities and saves the bill. It also writes a plain-text invoice to a .txt file using a character-based file writer. We hit a problem during testing where the invoice used line-drawing characters that displayed correctly on our machines but came out as garbled symbols on Windows computers set to a different text encoding. We fixed it by switching to plain dashes and vertical bars instead, which encode the same in everything.

Code for BillingService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Bill;
import com.pharmacy.model.BillItem;
import com.pharmacy.util.DataStore;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;
/**
* Manages billing and invoice operations.
* Demonstrates: ArrayList, Serialization, FileWriter, Method overloading.
*/
public class BillingService {
private ArrayList<Bill> billList;
private DataStore<Bill> dataStore;
private InventoryManager inventoryManager;
private static final String DATA_FILE = "data/bills.dat";
private static final String BACKUP_FILE = "data/backup_bills.txt";
private static final String INVOICE_DIR = "data/invoices/";
public BillingService(InventoryManager inventoryManager) {
this.inventoryManager = inventoryManager;
this.dataStore = new DataStore<>(DATA_FILE);
this.billList = dataStore.loadAll();
}
// Create and save a bill
public synchronized boolean createBill(Bill bill) {
// Reduce stock for each item
for (BillItem item : bill.getItems()) {
boolean success = inventoryManager.reduceStock(item.getMedicineId(), item.getQuantity());
if (!success) {
return false; // Insufficient stock
}
}
billList.add(bill);
saveData();
generateInvoice(bill);
return true;
}
/**
* Generate invoice as text file.
* Demonstrates: FileWriter usage.
*/
public void generateInvoice(Bill bill) {
java.io.File dir = new java.io.File(INVOICE_DIR);
if (!dir.exists()) {
dir.mkdirs();
}
String invoicePath = INVOICE_DIR + bill.getBillId() + "_invoice.txt";
try (FileWriter writer = new FileWriter(invoicePath)) {
```

```
writer.write("■■■
```

```
total += bill.getNetAmount();
}
}
return total;
}
private void saveData() {
dataStore.saveAll(billList);
}
public void createBackup() {
dataStore.backupToText(billList, BACKUP_FILE);
}
public int getBillCount() {
return billList.size();
}
}
```

13. CustomerService.java

Manages patient records: add, update, delete, search, and fetch by ID. The storage structure is a list and a map in parallel, same as InventoryManager. The list is used when displaying all customers in a table. The map is used when the billing screen or prescription screen needs a specific customer by their ID without scanning the whole list. The search method does a case-insensitive name match, returning every customer whose name contains the search text anywhere within it. Every change is written to disk immediately after it happens.

Code for CustomerService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Customer;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages customer/patient records.
* Demonstrates: ArrayList, HashMap, Iterator, Generics.
*/
public class CustomerService {
private ArrayList<Customer> customerList;
private HashMap<String, Customer> customerMap;
private DataStore<Customer> dataStore;
private static final String DATA_FILE = "data/customers.dat";
private static final String BACKUP_FILE = "data/backup_customers.txt";
public CustomerService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.customerList = dataStore.loadAll();
this.customerMap = new HashMap<>();
for (Customer cust : customerList) {
customerMap.put(cust.getCustomerId(), cust);
}
}
public synchronized void addCustomer(Customer customer) {
customerList.add(customer);
customerMap.put(customer.getCustomerId(), customer);
saveData();
}
```

```
public synchronized void updateCustomer(Customer customer) {
customerMap.put(customer.getCustomerId(), customer);
for (int i = 0; i < customerList.size(); i++) {
if (customerList.get(i).getCustomerId().equals(customer.getCustomerId())) {
customerList.set(i, customer);
break;
}
}
saveData();
}
public synchronized void deleteCustomer(String customerId) {
Iterator<Customer> iterator = customerList.iterator();
while (iterator.hasNext()) {
Customer cust = iterator.next();
if (cust.getCustomerId().equals(customerId)) {
iterator.remove();
break;
}
}
customerMap.remove(customerId);
saveData();
}
public Customer getCustomerById(String customerId) {
return customerMap.get(customerId);
}
public ArrayList<Customer> searchByName(String name) {
ArrayList<Customer> results = new ArrayList<>();
for (Customer cust : customerList) {
if (cust.getName().toLowerCase().contains(name.toLowerCase())) {
results.add(cust);
}
}
return results;
}
public ArrayList<Customer> getAllCustomers() {
return customerList;
}
private void saveData() {
dataStore.saveAll(customerList);
}
public void createBackup() {
dataStore.backupToText(customerList, BACKUP_FILE);
}
public int getCustomerCount() {
return customerList.size();
}
}
```

14. SupplierService.java

Handles supplier records with the same structure as CustomerService. List plus map, iterator-based deletion, immediate persistence after each change. The only difference is that search runs against the company name field rather than a person's name. This class is also called by the backup service, which asks it to write a readable text copy of all supplier data to a separate file.

Code for SupplierService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Supplier;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Iterator;
/**
* Manages supplier records.
* Demonstrates: ArrayList, HashMap, Iterator, Generics.
*/
public class SupplierService {
private ArrayList<Supplier> supplierList;
private HashMap<String, Supplier> supplierMap;
private DataStore<Supplier> dataStore;
private static final String DATA_FILE = "data/suppliers.dat";
private static final String BACKUP_FILE = "data/backup_suppliers.txt";
public SupplierService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.supplierList = dataStore.loadAll();
this.supplierMap = new HashMap<>();
for (Supplier sup : supplierList) {
supplierMap.put(sup.getSupplierId(), sup);
}
}
public synchronized void addSupplier(Supplier supplier) {
supplierList.add(supplier);
supplierMap.put(supplier.getSupplierId(), supplier);
saveData();
}
public synchronized void updateSupplier(Supplier supplier) {
supplierMap.put(supplier.getSupplierId(), supplier);
for (int i = 0; i < supplierList.size(); i++) {
if (supplierList.get(i).getSupplierId().equals(supplier.getSupplierId())) {
supplierList.set(i, supplier);
break;
}
}
saveData();
}
public synchronized void deleteSupplier(String supplierId) {
Iterator<Supplier> iterator = supplierList.iterator();
while (iterator.hasNext()) {
Supplier sup = iterator.next();
if (sup.getSupplierId().equals(supplierId)) {
iterator.remove();
break;
}
}
supplierMap.remove(supplierId);
saveData();
}
public Supplier getSupplierById(String supplierId) {
return supplierMap.get(supplierId);
}
public ArrayList<Supplier> searchByName(String name) {
```

```
ArrayList<Supplier> results = new ArrayList<>();
for (Supplier sup : supplierList) {
if (sup.getCompanyName().toLowerCase().contains(name.toLowerCase())) {
results.add(sup);
}
}
return results;
}
public ArrayList<Supplier> getAllSuppliers() {
return supplierList;
}
private void saveData() {
dataStore.saveAll(supplierList);
}
public void createBackup() {
dataStore.backupToText(supplierList, BACKUP_FILE);
}
public int getSupplierCount() {
return supplierList.size();
}
}
```

15. PrescriptionService.java

Prescriptions can be added, deleted, and retrieved here. One method worth mentioning is getByCustomerId, which returns all prescriptions belonging to a specific patient. The prescription screen uses this when a staff member selects a patient to see their prescription history. The persistence mechanism is identical to the other service classes - a DataStore instance handles reading and writing the serialised file.

Code for PrescriptionService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Prescription;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
import java.util.HashMap;
/**
* Manages prescription records.
* Demonstrates: ArrayList, HashMap, Generics, Serialization.
*/
public class PrescriptionService {
private ArrayList<Prescription> prescriptionList;
private HashMap<String, Prescription> prescriptionMap;
private DataStore<Prescription> dataStore;
private static final String DATA_FILE = "data/prescriptions.dat";
private static final String BACKUP_FILE = "data/backup_prescriptions.txt";
public PrescriptionService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.prescriptionList = dataStore.loadAll();
this.prescriptionMap = new HashMap<>();
for (Prescription pres : prescriptionList) {
prescriptionMap.put(pres.getPrescriptionId(), pres);
}
}
public synchronized void addPrescription(Prescription prescription) {
```

```
prescriptionList.add(prescription);
prescriptionMap.put(prescription.getPrescriptionId(), prescription);
saveData();
}
public synchronized void deletePrescription(String prescriptionId) {
prescriptionList.removeIf(p -> p.getPrescriptionId().equals(prescriptionId));
prescriptionMap.remove(prescriptionId);
saveData();
}
public Prescription getPrescriptionById(String prescriptionId) {
return prescriptionMap.get(prescriptionId);
}
public ArrayList<Prescription> getByCustomerId(String customerId) {
ArrayList<Prescription> results = new ArrayList<>();
for (Prescription pres : prescriptionList) {
if (pres.getCustomerId().equals(customerId)) {
results.add(pres);
}
}
return results;
}
public ArrayList<Prescription> getAllPrescriptions() {
return prescriptionList;
}
private void saveData() {
dataStore.saveAll(prescriptionList);
}
public void createBackup() {
dataStore.backupToText(prescriptionList, BACKUP_FILE);
}
public int getPrescriptionCount() {
return prescriptionList.size();
}
}
```

16. UserService.java

This class handles login and user account management. On first run, when the data folder has no user file yet, it creates two default accounts: one admin and one pharmacist. The login process searches the user list for a matching username, then calls the authenticate method on the User object to check the password. We made that check a method on User rather than in this service class so the password comparison logic is in exactly one place and is not duplicated if we add a password-change feature later.

Code for UserService.java:

```
package com.pharmacy.service;
import com.pharmacy.model.User;
import com.pharmacy.model.UserRole;
import com.pharmacy.util.DataStore;
import java.util.ArrayList;
/**
* Manages user authentication and accounts.
* Demonstrates: ArrayList, Serialization, role-based access.
*/
public class UserService {
```

```
private ArrayList<User> userList;
private DataStore<User> dataStore;
private User currentUser;
private static final String DATA_FILE = "data/users.dat";
public UserService() {
this.dataStore = new DataStore<>(DATA_FILE);
this.userList = dataStore.loadAll();
// Create default admin if no users exist
if (userList.isEmpty()) {
createDefaultUsers();
}
}
// Create default users for first run
private void createDefaultUsers() {
userList.add(new User("admin", "admin123", "System Administrator", UserRole.ADMIN));
userList.add(new User("pharmacist", "pharma123", "Default Pharmacist", UserRole.PHARMACIST));
dataStore.saveAll(userList);
}
// Authenticate user
public User authenticate(String username, String password) {
for (User user : userList) {
if (user.getUsername().equals(username) && user.authenticate(password)) {
this.currentUser = user;
return user;
}
}
return null;
}
public User getCurrentUser() {
return currentUser;
}
public void setCurrentUser(User user) {
this.currentUser = user;
}
public void addUser(User user) {
userList.add(user);
dataStore.saveAll(userList);
}
public ArrayList<User> getAllUsers() {
return userList;
}
public void logout() {
this.currentUser = null;
}
}
```

17. ExpiryAlertThread.java

The expiry monitor runs as a background thread by directly extending the Thread class. It is configured as a daemon thread, which means the JVM will shut it down automatically when the application closes without needing any manual stop call. The run loop sleeps for thirty seconds, then wakes up and asks the inventory manager for two lists: medicines already past their expiry date and medicines expiring within the next thirty days. A wait/notify mechanism is used so that any external code that wants to read those lists can safely wait until the thread has finished its current check before accessing the results.

Code for ExpiryAlertThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.Date;
/**
* Background thread that checks for expired and soon-to-expire medicines.
* Demonstrates: Thread class, sleep(), join(), synchronized, Runnable.
*/
public class ExpiryAlertThread extends Thread {
private InventoryManager inventoryManager;
private volatile boolean running = true;
private ArrayList<Medicine> expiredMedicines;
private ArrayList<Medicine> expiringMedicines;
private final Object lock = new Object();
private AlertListener listener;
// Interface for callback (Abstraction demonstration)
public interface AlertListener {
void onAlertUpdate(ArrayList<Medicine> expired, ArrayList<Medicine> expiringSoon);
}
public ExpiryAlertThread(InventoryManager inventoryManager) {
super("ExpiryAlertThread");
this.inventoryManager = inventoryManager;
this.expiredMedicines = new ArrayList<>();
this.expiringMedicines = new ArrayList<>();
setDaemon(true); // Daemon thread - will stop when main thread stops
}
public void setAlertListener(AlertListener listener) {
this.listener = listener;
}
@Override
public void run() {
while (running) {
try {
checkExpiry();
// Sleep for 30 seconds between checks
Thread.sleep(30000);
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
break;
}
}
}
// Synchronized method for thread-safe expiry checking
private synchronized void checkExpiry() {
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
String currentDate = sdf.format(new Date());
// Calculate date 30 days from now
Calendar cal = Calendar.getInstance();
cal.add(Calendar.DAY_OF_MONTH, 30);
String thresholdDate = sdf.format(cal.getTime());
synchronized (lock) {
```

```
expiredMedicines = inventoryManager.getExpiredMedicines(currentDate);
expiringMedicines = inventoryManager.getExpiringSoon(currentDate, thresholdDate);
// Notify waiting threads using wait/notify mechanism
lock.notifyAll();
}
// Callback to listener
if (listener != null) {
listener.onAlertUpdate(expiredMedicines, expiringMedicines);
}
}
// Method for other threads to wait for alert data
public ArrayList<Medicine> waitForExpiredList() throws InterruptedException {
synchronized (lock) {
lock.wait(5000); // Wait up to 5 seconds
return new ArrayList<>(expiredMedicines);
}
}
public ArrayList<Medicine> getExpiredMedicines() {
synchronized (lock) {
return new ArrayList<>(expiredMedicines);
}
}
public ArrayList<Medicine> getExpiringMedicines() {
synchronized (lock) {
return new ArrayList<>(expiringMedicines);
}
}
public void stopChecking() {
running = false;
this.interrupt();
}
}
```

18. StockAlertThread.java

The stock monitor uses the Runnable interface rather than extending Thread directly. This was deliberate - we wanted to show both approaches in the same project since the syllabus covers both. The thread runs a loop that pauses for thirty seconds at a time, then calls the inventory manager for a list of any medicines below ten units. A volatile boolean controls whether the loop continues. It is volatile specifically so that when the main thread sets it to false to stop the worker, the worker thread sees the change immediately rather than reading a cached copy of the old value.

Code for StockAlertThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Medicine;
import java.util.ArrayList;
/**
* Background thread that monitors low stock levels.
* Demonstrates: Runnable interface, Thread, sleep(), synchronized.
*/
public class StockAlertThread implements Runnable {
private InventoryManager inventoryManager;
private volatile boolean running = true;
private ArrayList<Medicine> lowStockMedicines;
private AlertListener listener;
```

```
// Callback interface (Abstraction)
public interface AlertListener {
void onLowStockUpdate(ArrayList<Medicine> lowStockMedicines);
}
public StockAlertThread(InventoryManager inventoryManager) {
this.inventoryManager = inventoryManager;
this.lowStockMedicines = new ArrayList<>();
}
public void setAlertListener(AlertListener listener) {
this.listener = listener;
}
@Override
public void run() {
while (running) {
try {
checkStock();
Thread.sleep(30000); // Check every 30 seconds
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
break;
}
}
}
private synchronized void checkStock() {
lowStockMedicines = inventoryManager.getLowStockMedicines();
if (listener != null) {
listener.onLowStockUpdate(lowStockMedicines);
}
}
public synchronized ArrayList<Medicine> getLowStockMedicines() {
return new ArrayList<>(lowStockMedicines);
}
public void stopChecking() {
running = false;
}
}
```

19. BillingThread.java

When the cashier clicks "Generate Bill", the actual processing - stock checks, quantity deduction, file writing, record saving - runs here rather than on the main application thread. Running it on the main thread would freeze the entire interface for a second or two. This thread does the work, then needs to update status labels in the UI when done. UI updates must happen on the JavaFX main thread, so the result is passed to a scheduling method that queues the update to run on the correct thread as soon as it is available. Separately, a small watcher thread calls join on this billing thread, which demonstrates how one thread can pause and wait for another to complete before continuing.

Code for BillingThread.java:

```
package com.pharmacy.service;
import com.pharmacy.model.Bill;
/**
* Multi-threaded billing simulation.
* Demonstrates: Thread, Runnable, synchronized, sleep(), join().
*
* Simulates billing processing in a separate thread to keep UI responsive.
```

```
*/
public class BillingThread extends Thread {
private BillingService billingService;
private Bill bill;
private boolean success;
private String message;
private BillingCallback callback;
// Callback interface for billing result (Abstraction)
public interface BillingCallback {
void onBillingComplete(boolean success, String message, Bill bill);
}
public BillingThread(BillingService billingService, Bill bill, BillingCallback callback) {
super("BillingThread-" + bill.getBillId());
this.billingService = billingService;
this.bill = bill;
this.callback = callback;
this.success = false;
this.message = "";
}
@Override
public void run() {
try {
// Simulate processing time
System.out.println("[" + getName() + "] Processing bill " + bill.getBillId() + "...");
Thread.sleep(1000); // Simulate processing delay
// Process the bill
synchronized (billingService) {
success = billingService.createBill(bill);
}
if (success) {
message = "Bill " + bill.getBillId() + " processed successfully!";
System.out.println("[" + getName() + "] " + message);
} else {
message = "Failed to process bill " + bill.getBillId() + ". Insufficient stock.";
System.out.println("[" + getName() + "] " + message);
}
// Simulate invoice generation time
Thread.sleep(500);
} catch (InterruptedException e) {
message = "Billing interrupted for " + bill.getBillId();
Thread.currentThread().interrupt();
}
// Callback
if (callback != null) {
callback.onBillingComplete(success, message, bill);
}
}
public boolean isSuccess() {
return success;
}
public String getMessage() {
return message;
}
}
```

20. BackupService.java

This class implements the Runnable interface and is designed to be passed to whichever Thread the caller creates for it. When it executes, it calls the backup method on each of the five service classes in order - inventory, customers, suppliers, bills, prescriptions - with a short pause between each. Each service class writes its data out as a human-readable text file in the data folder. A callback interface carries the completion message back to the UI, which shows it to the user once the backup finishes.

Code for BackupService.java:

```
package com.pharmacy.service;
/**
* Backup service that creates file backups of all data.
* Demonstrates: Runnable, Thread, FileWriter (via DataStore).
*/
public class BackupService implements Runnable {
private InventoryManager inventoryManager;
private CustomerService customerService;
private SupplierService supplierService;
private BillingService billingService;
private PrescriptionService prescriptionService;
private BackupCallback callback;
public interface BackupCallback {
void onBackupComplete(String message);
}
public BackupService(InventoryManager inventoryManager, CustomerService customerService,
SupplierService supplierService, BillingService billingService,
PrescriptionService prescriptionService) {
this.inventoryManager = inventoryManager;
this.customerService = customerService;
this.supplierService = supplierService;
this.billingService = billingService;
this.prescriptionService = prescriptionService;
}
public void setCallback(BackupCallback callback) {
this.callback = callback;
}
@Override
public void run() {
try {
System.out.println("[BackupService] Starting backup...");
Thread.sleep(500);
inventoryManager.createBackup();
Thread.sleep(200);
customerService.createBackup();
Thread.sleep(200);
supplierService.createBackup();
Thread.sleep(200);
billingService.createBackup();
Thread.sleep(200);
prescriptionService.createBackup();
Thread.sleep(200);
String msg = "All data backed up successfully!";
System.out.println("[BackupService] " + msg);
if (callback != null) {
```

```
callback.onBackupComplete(msg);
}
} catch (InterruptedException e) {
Thread.currentThread().interrupt();
if (callback != null) {
callback.onBackupComplete("Backup interrupted!");
}
}
}
}
```

UI Package (com.pharmacy.ui)

21. PharmacyApp.java

This is the main class where execution begins. It extends Application, which is the base class JavaFX requires as the entry point. The start method, which JavaFX calls automatically, creates all the service objects, starts the two background alert threads, loads sample data if the data folder is empty, and shows the login screen. Navigation between screens is handled through dedicated show methods - showInventoryScreen, showBillingScreen, and so on - each of which creates the target screen object and calls setScene on the application window to replace what is currently visible.

Code for PharmacyApp.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.User;
import com.pharmacy.service.*;
import com.pharmacy.util.SampleDataLoader;
import javafx.application.Application;
import javafx.application.Platform;
import javafx.stage.Stage;
import java.io.File;
/**
* Main JavaFX Application class - Entry point for the Pharmacy Management System.
* Demonstrates: JavaFX Application, Stage management, Inheritance (extends Application).
*/
public class PharmacyApp extends Application {
// Service instances (shared across all screens)
private UserService userService;
private InventoryManager inventoryManager;
private CustomerService customerService;
private SupplierService supplierService;
private BillingService billingService;
private PrescriptionService prescriptionService;
// Alert threads
private ExpiryAlertThread expiryAlertThread;
private Thread stockAlertThreadHandle;
private StockAlertThread stockAlertRunnable;
private Stage primaryStage;
@Override
public void start(Stage primaryStage) {
this.primaryStage = primaryStage;
// Create data directory
new File("data/invoices").mkdirs();
// Initialize all services
```

```
userService = new UserService();
inventoryManager = new InventoryManager();
customerService = new CustomerService();
supplierService = new SupplierService();
billingService = new BillingService(inventoryManager);
prescriptionService = new PrescriptionService();
// Load sample data if empty
SampleDataLoader.loadSampleData(inventoryManager, customerService, supplierService);
// Start background alert threads (Multithreading demonstration)
startAlertThreads();
// Show login screen
showLoginScreen();
primaryStage.setTitle("PharmaCare - Pharmacy Management System");
primaryStage.setWidth(1100);
primaryStage.setHeight(700);
primaryStage.setOnCloseRequest(e -> {
stopAlertThreads();
Platform.exit();
System.exit(0);
});
primaryStage.show();
}
/**
* Start background threads for monitoring.
* Demonstrates: Thread creation (both Thread subclass and Runnable), start(), daemon threads.
*/
private void startAlertThreads() {
// Thread extending Thread class
expiryAlertThread = new ExpiryAlertThread(inventoryManager);
expiryAlertThread.start();
// Thread using Runnable interface
stockAlertRunnable = new StockAlertThread(inventoryManager);
stockAlertThreadHandle = new Thread(stockAlertRunnable, "StockAlertThread");
stockAlertThreadHandle.setDaemon(true);
stockAlertThreadHandle.start();
}
private void stopAlertThreads() {
if (expiryAlertThread != null) {
expiryAlertThread.stopChecking();
}
if (stockAlertRunnable != null) {
stockAlertRunnable.stopChecking();
}
}
// Show login screen
public void showLoginScreen() {
LoginScreen loginScreen = new LoginScreen(this, userService);
primaryStage.setScene(loginScreen.getScene());
}
// Show dashboard after login
public void showDashboard(User user) {
DashboardScreen dashboard = new DashboardScreen(this, user);
primaryStage.setScene(dashboard.getScene());
}
// Show inventory module
```

```
public void showInventoryScreen() {
InventoryScreen screen = new InventoryScreen(this, inventoryManager, supplierService);
primaryStage.setScene(screen.getScene());
}
// Show billing module
public void showBillingScreen() {
BillingScreen screen = new BillingScreen(this, billingService, inventoryManager,
customerService, userService);
primaryStage.setScene(screen.getScene());
}
// Show supplier module
public void showSupplierScreen() {
SupplierScreen screen = new SupplierScreen(this, supplierService);
primaryStage.setScene(screen.getScene());
}
// Show customer module
public void showCustomerScreen() {
CustomerScreen screen = new CustomerScreen(this, customerService);
primaryStage.setScene(screen.getScene());
}
// Show prescription module
public void showPrescriptionScreen() {
PrescriptionScreen screen = new PrescriptionScreen(this, prescriptionService,
customerService, inventoryManager);
primaryStage.setScene(screen.getScene());
}
// Show reports module
public void showReportsScreen() {
ReportsScreen screen = new ReportsScreen(this, inventoryManager, billingService,
customerService, supplierService,
expiryAlertThread, stockAlertRunnable);
primaryStage.setScene(screen.getScene());
}
// Getters for services
public UserService getUserService() { return userService; }
public InventoryManager getInventoryManager() { return inventoryManager; }
public CustomerService getCustomerService() { return customerService; }
public SupplierService getSupplierService() { return supplierService; }
public BillingService getBillingService() { return billingService; }
public PrescriptionService getPrescriptionService() { return prescriptionService; }
public ExpiryAlertThread getExpiryAlertThread() { return expiryAlertThread; }
public StockAlertThread getStockAlertRunnable() { return stockAlertRunnable; }
public Stage getPrimaryStage() { return primaryStage; }
/**
* Main method - entry point.
*/
public static void main(String[] args) {
launch(args);
}
}
```

22. LoginScreen.java

The login form is built using a grid layout that keeps the labels and fields neatly aligned. The password field uses JavaFX's PasswordField component, which shows dots instead of the actual characters. When the user clicks Login, the username and password are passed to UserService. If

authentication fails, a red label appears below the button explaining why. If it succeeds, the authenticated user object is stored in the service and the dashboard screen is loaded. The role stored in that user object shapes what options are available for the rest of the session.

Code for LoginScreen.java:

```
package com.pharmacy.ui;

import com.pharmacy.model.User;
import com.pharmacy.service.UserService;

import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;

/**
* Login Screen with role-based authentication.
* Demonstrates: JavaFX (GridPane, TextField, Button, Label), Event handling.
*/
public class LoginScreen {

private Scene scene;
private PharmacyApp app;
private UserService userService;

public LoginScreen(PharmacyApp app, UserService userService) {
this.app = app;
this.userService = userService;
createScene();
}

private void createScene() {
// Main layout
VBox mainLayout = new VBox(20);
mainLayout.setAlignment(Pos.CENTER);
mainLayout.setPadding(new Insets(50));
mainLayout.setStyle("-fx-background-color: #2c3e50;");

// Title
Label titleLabel = new Label("PharmaCare");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 36));
titleLabel.setStyle("-fx-text-fill: #ecf0f1;");

Label subtitleLabel = new Label("Pharmacy Management System");
subtitleLabel.setFont(Font.font("Arial", FontWeight.NORMAL, 16));
subtitleLabel.setStyle("-fx-text-fill: #bdc3c7;");

// Login form using GridPane
GridPane grid = new GridPane();
grid.setAlignment(Pos.CENTER);
grid.setHgap(10);
grid.setVgap(15);
grid.setPadding(new Insets(30));
grid.setMaxWidth(400);
grid.setStyle("-fx-background-color: #34495e; -fx-background-radius: 10;");

Label loginTitle = new Label("Login");
loginTitle.setFont(Font.font("Arial", FontWeight.BOLD, 20));
loginTitle.setStyle("-fx-text-fill: #ecf0f1;");
grid.add(loginTitle, 0, 0, 2, 1);

Label userLabel = new Label("Username:");
userLabel.setStyle("-fx-text-fill: #ecf0f1;");
```

```
TextField usernameField = new TextField();
usernameField.setPromptText("Enter username");
usernameField.setPrefWidth(250);
grid.add(userLabel, 0, 1);
grid.add(usernameField, 1, 1);
Label passLabel = new Label("Password:");
passLabel.setStyle("-fx-text-fill: #ecf0f1;");
PasswordField passwordField = new PasswordField();
passwordField.setPromptText("Enter password");
grid.add(passLabel, 0, 2);
grid.add(passwordField, 1, 2);
Label messageLabel = new Label();
messageLabel.setStyle("-fx-text-fill: #e74c3c;");
Button loginBtn = new Button("Login");
loginBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; " +
"-fx-font-size: 14; -fx-pref-width: 250; -fx-cursor: hand;");
loginBtn.setOnAction(e -> {
String username = usernameField.getText().trim();
String password = passwordField.getText().trim();
if (username.isEmpty() || password.isEmpty()) {
messageLabel.setText("Please enter both username and password.");
return;
}
User user = userService.authenticate(username, password);
if (user != null) {
app.showDashboard(user);
} else {
messageLabel.setText("Invalid credentials! Please try again.");
passwordField.clear();
}
});
// Allow Enter key to login
passwordField.setOnAction(e -> loginBtn.fire());
usernameField.setOnAction(e -> loginBtn.fire());
grid.add(loginBtn, 0, 3, 2, 1);
grid.add(messageLabel, 0, 4, 2, 1);
// Default credentials hint
Label hintLabel = new Label("Default: admin/admin123 or pharmacist/pharma123");
hintLabel.setStyle("-fx-text-fill: #7f8c8d; -fx-font-size: 11;");
mainLayout.getChildren().addAll(titleLabel, subtitleLabel, grid, hintLabel);
scene = new Scene(mainLayout, 1100, 700);
}
public Scene getScene() {
return scene;
}
}
```

23. DashboardScreen.java

After logging in successfully, the dashboard is the first thing the user sees. The layout has a dark-coloured header bar across the top containing the application name and the logged-in user's name and role. Below that, six equal-sized buttons are arranged in two rows of three, each one opening a different module. A Logout button in the header ends the session by clearing the current user in UserService and returning to the login screen.

Code for DashboardScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.User;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Dashboard screen - main navigation hub after login.
* Demonstrates: JavaFX (VBox, HBox, Button, Label, GridPane), Event handling.
*/
public class DashboardScreen {
private Scene scene;
private PharmacyApp app;
private User currentUser;
public DashboardScreen(PharmacyApp app, User currentUser) {
this.app = app;
this.currentUser = currentUser;
createScene();
}
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(20);
topBar.setPadding(new Insets(15, 25, 15, 25));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Label titleLabel = new Label("PharmaCare Dashboard");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 22));
titleLabel.setStyle("-fx-text-fill: white;");
Region spacer = new Region();
HBox.setHgrow(spacer, Priority.ALWAYS);
Label userLabel = new Label("Welcome, " + currentUser.getFullName()
+ " (" + currentUser.getRole().getDisplayName() + ")");
userLabel.setStyle("-fx-text-fill: #bdc3c7; -fx-font-size: 13;");
Button logoutBtn = new Button("Logout");
logoutBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
logoutBtn.setOnAction(e -> {
app.getUserService().logout();
app.showLoginScreen();
});
topBar.getChildren().addAll(titleLabel, spacer, userLabel, logoutBtn);
root.setTop(topBar);
// Center - Module buttons in a grid
GridPane moduleGrid = new GridPane();
moduleGrid.setAlignment(Pos.CENTER);
moduleGrid.setHgap(25);
moduleGrid.setVgap(25);
moduleGrid.setPadding(new Insets(40));
// Module buttons
```

```
Button inventoryBtn = createModuleButton("Inventory\nManagement",
"Manage medicines, stock,\nbatch & expiry tracking", "#3498db");
inventoryBtn.setOnAction(e -> app.showInventoryScreen());
Button billingBtn = createModuleButton("Sales &\nBilling",
"Create bills, generate\ninvoices for customers", "#27ae60");
billingBtn.setOnAction(e -> app.showBillingScreen());
Button supplierBtn = createModuleButton("Supplier\nManagement",
"Manage supplier records\nand contact details", "#e67e22");
supplierBtn.setOnAction(e -> app.showSupplierScreen());
Button customerBtn = createModuleButton("Customer\nRecords",
"Manage patient/customer\ninformation", "#9b59b6");
customerBtn.setOnAction(e -> app.showCustomerScreen());
Button prescriptionBtn = createModuleButton("Prescription\nHandling",
"Record and manage\nprescription details", "#1abc9c");
prescriptionBtn.setOnAction(e -> app.showPrescriptionScreen());
Button reportsBtn = createModuleButton("Reports &\nAlerts",
"View sales reports, stock\n& expiry alerts", "#e74c3c");
reportsBtn.setOnAction(e -> app.showReportsScreen());
// Arrange in 3x2 grid
moduleGrid.add(inventoryBtn, 0, 0);
moduleGrid.add(billingBtn, 1, 0);
moduleGrid.add(supplierBtn, 2, 0);
moduleGrid.add(customerBtn, 0, 1);
moduleGrid.add(prescriptionBtn, 1, 1);
moduleGrid.add(reportsBtn, 2, 1);
root.setCenter(moduleGrid);
// Bottom status bar
HBox statusBar = new HBox(20);
statusBar.setPadding(new Insets(10, 25, 10, 25));
statusBar.setStyle("-fx-background-color: #34495e;");
Label statusLabel = new Label("Medicines: " + app.getInventoryManager().getMedicineCount()
+ " | Customers: " + app.getCustomerService().getCustomerCount()
+ " | Suppliers: " + app.getSupplierService().getSupplierCount()
+ " | Bills: " + app.getBillingService().getBillCount());
statusLabel.setStyle("-fx-text-fill: #bdc3c7; -fx-font-size: 12;");
statusBar.getChildren().add(statusLabel);
root.setBottom(statusBar);
scene = new Scene(root, 1100, 700);
}
// Helper to create styled module buttons
private Button createModuleButton(String title, String description, String color) {
VBox content = new VBox(8);
content.setAlignment(Pos.CENTER);
Label titleLbl = new Label(title);
titleLbl.setFont(Font.font("Arial", FontWeight.BOLD, 16));
titleLbl.setStyle("-fx-text-fill: white; -fx-text-alignment: center;");
Label descLbl = new Label(description);
descLbl.setStyle("-fx-text-fill: #ecf0f1; -fx-font-size: 11; -fx-text-alignment: center;");
content.getChildren().addAll(titleLbl, descLbl);
Button btn = new Button();
btn.setGraphic(content);
btn.setPrefSize(230, 130);
btn.setStyle("-fx-background-color: " + color + "; -fx-background-radius: 10; -fx-cursor:
hand;");
```

```
return btn;
}
public Scene getScene() {
return scene;
}
}
```

24. InventoryScreen.java

The inventory screen is the most visually complex in the system. A large table in the centre shows all medicines currently in stock, with columns for ID, name, category, batch, expiry date, quantity, and price. Clicking a row fills the form on the left side with that medicine's details so staff can edit and save. Above the table there are three controls: a search box that filters by medicine name as you type, a dropdown to sort the table by name, price, or expiry date, and a category dropdown that shows only medicines matching the selected category.

Code for InventoryScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Medicine;
import com.pharmacy.model.MedicineCategory;
import com.pharmacy.service.InventoryManager;
import com.pharmacy.service.SupplierService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.util.ArrayList;
/**
* Inventory Management Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Buttons, ComboBox.
*/
public class InventoryScreen {
private Scene scene;
private PharmacyApp app;
private InventoryManager inventoryManager;
private SupplierService supplierService;
private TableView<Medicine> tableView;
private ObservableList<Medicine> tableData;
public InventoryScreen(PharmacyApp app, InventoryManager inventoryManager,
SupplierService supplierService) {
this.app = app;
this.inventoryManager = inventoryManager;
this.supplierService = supplierService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
```

```
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = createTopBar("Inventory Management");
root.setTop(topBar);
// Left side - Form
VBox formBox = createForm();
root.setLeft(formBox);
// Center - Table
VBox tableBox = createTable();
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private HBox createTopBar(String title) {
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label(title);
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
return topBar;
}
private VBox createForm() {
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("Add / Edit Medicine");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
// Form fields using GridPane
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");
TextField nameField = new TextField();
nameField.setPromptText("Medicine name");
TextField mfgField = new TextField();
mfgField.setPromptText("Manufacturer");
ComboBox<MedicineCategory> categoryBox = new ComboBox<>();
categoryBox.getItems().addAll(MedicineCategory.values());
categoryBox.setPromptText("Select Category");
categoryBox.setPrefWidth(180);
TextField batchField = new TextField();
batchField.setPromptText("Batch number");
TextField expiryField = new TextField();
expiryField.setPromptText("YYYY-MM-DD");
TextField qtyField = new TextField();
```

```
qtyField.setPromptText("Quantity");
TextField priceField = new TextField();
priceField.setPromptText("Price (Rs.)");
TextField supplierField = new TextField();
supplierField.setPromptText("Supplier ID");
int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Name:"), 0, row); grid.add(nameField, 1, row++);
grid.add(new Label("Manufacturer:"), 0, row); grid.add(mfgField, 1, row++);
grid.add(new Label("Category:"), 0, row); grid.add(categoryBox, 1, row++);
grid.add(new Label("Batch No:"), 0, row); grid.add(batchField, 1, row++);
grid.add(new Label("Expiry Date:"), 0, row); grid.add(expiryField, 1, row++);
grid.add(new Label("Quantity:"), 0, row); grid.add(qtyField, 1, row++);
grid.add(new Label("Price:"), 0, row); grid.add(priceField, 1, row++);
grid.add(new Label("Supplier ID:"), 0, row); grid.add(supplierField, 1, row++);
Label statusLabel = new Label();
statusLabel.setStyle("-fx-text-fill: #27ae60;");
// Buttons
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
// Add button action
addBtn.setOnAction(e -> {
try {
String id = IDGenerator.generateMedicineId();
Medicine med = new Medicine(id, nameField.getText(), mfgField.getText(),
categoryBox.getValue(), batchField.getText(), expiryField.getText(),
Integer.parseInt(qtyField.getText()), // Wrapper class parsing
Double.parseDouble(priceField.getText()), // Wrapper class parsing
supplierField.getText());
inventoryManager.addMedicine(med);
refreshTable();
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("Medicine added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Update button action
updateBtn.setOnAction(e -> {
try {
String id = idField.getText();
if (id.isEmpty()) {
statusLabel.setText("Select a medicine first!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
```

```
}
Medicine med = new Medicine(id, nameField.getText(), mfgField.getText(),
categoryBox.getValue(), batchField.getText(), expiryField.getText(),
Integer.parseInt(qtyField.getText()),
Double.parseDouble(priceField.getText()),
supplierField.getText());
inventoryManager.updateMedicine(med);
refreshTable();
statusLabel.setText("Medicine updated: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Delete button action
deleteBtn.setOnAction(e -> {
String id = idField.getText();
if (!id.isEmpty()) {
inventoryManager.deleteMedicine(id);
refreshTable();
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("Medicine deleted: " + id);
}
});
// Clear button
clearBtn.setOnAction(e -> {
clearFields(idField, nameField, mfgField, categoryBox, batchField,
expiryField, qtyField, priceField, supplierField);
statusLabel.setText("");
});
// Table selection listener - populate form
tableView = new TableView<>();
tableView.getSelectionModel().selectedItemProperty().addListener((obs, oldVal, newVal) -> {
if (newVal != null) {
idField.setText(newVal.getMedicineId());
nameField.setText(newVal.getName());
mfgField.setText(newVal.getManufacturer());
categoryBox.setValue(newVal.getCategory());
batchField.setText(newVal.getBatchNumber());
expiryField.setText(newVal.getExpiryDate());
qtyField.setText(String.valueOf(newVal.getQuantity()));
priceField.setText(String.valueOf(newVal.getPrice()));
supplierField.setText(newVal.getSupplierId());
}
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
return formBox;
}
@SuppressWarnings("unchecked")
private VBox createTable() {
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
// Search and sort controls
HBox controlsBox = new HBox(10);
```

```
controlsBox.setAlignment(Pos.CENTER_LEFT);
TextField searchField = new TextField();
searchField.setPromptText("Search by name...");
searchField.setPrefWidth(200);
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String query = searchField.getText().trim();
if (query.isEmpty()) {
refreshTable();
} else {
ArrayList<Medicine> results = inventoryManager.searchByName(query);
tableData = FXCollections.observableArrayList(results);
tableView.setItems(tableData);
}
});
ComboBox<String> sortBox = new ComboBox<>();
sortBox.getItems().addAll("Sort by Name", "Sort by Price", "Sort by Expiry");
sortBox.setPromptText("Sort by...");
sortBox.setOnAction(e -> {
String selected = sortBox.getValue();
ArrayList<Medicine> sorted;
if ("Sort by Name".equals(selected)) {
sorted = inventoryManager.getSortedByName();
} else if ("Sort by Price".equals(selected)) {
sorted = inventoryManager.getSortedByPrice();
} else {
sorted = inventoryManager.getSortedByExpiry();
}
tableData = FXCollections.observableArrayList(sorted);
tableView.setItems(tableData);
});
ComboBox<MedicineCategory> filterBox = new ComboBox<>();
filterBox.getItems().addAll(MedicineCategory.values());
filterBox.setPromptText("Filter by Category");
filterBox.setOnAction(e -> {
MedicineCategory cat = filterBox.getValue();
if (cat != null) {
ArrayList<Medicine> filtered = inventoryManager.filterByCategory(cat);
tableData = FXCollections.observableArrayList(filtered);
tableView.setItems(tableData);
}
});
Button refreshBtn = new Button("Show All");
refreshBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
refreshBtn.setOnAction(e -> refreshTable());
controlsBox.getChildren().addAll(searchField, searchBtn, sortBox, filterBox, refreshBtn);
// Table columns
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
idCol.setPrefWidth(80);
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Medicine, MedicineCategory> catCol = new TableColumn<>("Category");
```

```
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
catCol.setPrefWidth(80);
TableColumn<Medicine, String> batchCol = new TableColumn<>("Batch");
batchCol.setCellValueFactory(new PropertyValueFactory<>("batchNumber"));
batchCol.setPrefWidth(70);
TableColumn<Medicine, String> expiryCol = new TableColumn<>("Expiry");
expiryCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expiryCol.setPrefWidth(90);
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
qtyCol.setPrefWidth(50);
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
priceCol.setPrefWidth(70);
tableView.getColumns().addAll(idCol, nameCol, catCol, batchCol, expiryCol, qtyCol, priceCol);
refreshTable();
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(controlsBox, tableView);
return tableBox;
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(inventoryManager.getAllMedicines());
tableView.setItems(tableData);
}
private void clearFields(TextField idField, TextField nameField, TextField mfgField,
ComboBox<MedicineCategory> categoryBox, TextField batchField,
TextField expiryField, TextField qtyField, TextField priceField,
TextField supplierField) {
idField.clear();
nameField.clear();
mfgField.clear();
categoryBox.setValue(null);
batchField.clear();
expiryField.clear();
qtyField.clear();
priceField.clear();
supplierField.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() {
return scene;
}
}
```

25. BillingScreen.java

The billing screen is divided into two sides. The left side is where a new bill is assembled: the staff member enters a customer ID (the name fills automatically from the customer database), then adds medicines one by one by entering the medicine ID and desired quantity. The running total updates after each item is added. An optional discount percentage can be applied before the final generate button is clicked. Once clicked, the bill goes to a BillingThread so the interface is not blocked during processing. The right side shows a table of current bill items and below it a table of all past bills.

Code for BillingScreen.java:

```
package com.pharmacy.ui;
```

```
import com.pharmacy.model.*;
import com.pharmacy.service.*;
import com.pharmacy.util.IDGenerator;
import javafx.application.Platform;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.Date;
/**
* Sales and Billing Screen with multi-threaded billing simulation.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Multithreaded billing.
*/
public class BillingScreen {
private Scene scene;
private PharmacyApp app;
private BillingService billingService;
private InventoryManager inventoryManager;
private CustomerService customerService;
private UserService userService;
private TableView<BillItem> billItemsTable;
private ObservableList<BillItem> billItems;
private Label totalLabel;
private Label netLabel;
private Bill currentBill;
public BillingScreen(PharmacyApp app, BillingService billingService,
InventoryManager inventoryManager, CustomerService customerService,
UserService userService) {
this.app = app;
this.billingService = billingService;
this.inventoryManager = inventoryManager;
this.customerService = customerService;
this.userService = userService;
this.billItems = FXCollections.observableArrayList();
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(userService.getCurrentUser()));
```

```
Label titleLabel = new Label("Sales & Billing");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Bill form
VBox formBox = new VBox(12);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("New Bill");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane custGrid = new GridPane();
custGrid.setVgap(8);
custGrid.setHgap(8);
TextField custIdField = new TextField();
custIdField.setPromptText("Customer ID");
TextField custNameField = new TextField();
custNameField.setPromptText("Customer Name");
// Auto-fill customer name when ID is entered
custIdField.textProperty().addListener((obs, oldVal, newVal) -> {
if (newVal != null && !newVal.isEmpty()) {
Customer cust = customerService.getCustomerById(newVal.trim());
if (cust != null) {
custNameField.setText(cust.getName());
}
}
});
custGrid.add(new Label("Customer ID:"), 0, 0);
custGrid.add(custIdField, 1, 0);
custGrid.add(new Label("Customer Name:"), 0, 1);
custGrid.add(custNameField, 1, 1);
// Add medicine to bill
Label addItemTitle = new Label("Add Medicine to Bill");
addItemTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
GridPane itemGrid = new GridPane();
itemGrid.setVgap(8);
itemGrid.setHgap(8);
TextField medIdField = new TextField();
medIdField.setPromptText("Medicine ID");
Label medNameLabel = new Label("-");
Label medPriceLabel = new Label("-");
Label medStockLabel = new Label("-");
TextField medQtyField = new TextField();
medQtyField.setPromptText("Quantity");
// Auto-fill medicine details
medIdField.textProperty().addListener((obs, oldVal, newVal) -> {
if (newVal != null && !newVal.isEmpty()) {
Medicine med = inventoryManager.getMedicineById(newVal.trim());
if (med != null) {
medNameLabel.setText(med.getName());
medPriceLabel.setText("Rs." + String.format("%.2f", med.getPrice()));
medStockLabel.setText("Stock: " + med.getQuantity());
}
```

```
}
});
int r = 0;
itemGrid.add(new Label("Medicine ID:"), 0, r); itemGrid.add(medIdField, 1, r++);
itemGrid.add(new Label("Name:"), 0, r); itemGrid.add(medNameLabel, 1, r++);
itemGrid.add(new Label("Price:"), 0, r); itemGrid.add(medPriceLabel, 1, r++);
itemGrid.add(new Label("Available:"), 0, r); itemGrid.add(medStockLabel, 1, r++);
itemGrid.add(new Label("Quantity:"), 0, r); itemGrid.add(medQtyField, 1, r++);
Label statusLabel = new Label();
Button addItemBtn = new Button("Add to Bill");
addItemBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;
-fx-pref-width: 200;");
addItemBtn.setOnAction(e -> {
try {
String medId = medIdField.getText().trim();
Medicine med = inventoryManager.getMedicineById(medId);
if (med == null) {
statusLabel.setText("Medicine not found!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
int qty = Integer.parseInt(medQtyField.getText().trim());
if (qty > med.getQuantity()) {
statusLabel.setText("Insufficient stock!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
BillItem item = new BillItem(medId, med.getName(), qty, med.getPrice());
billItems.add(item);
updateTotals();
medIdField.clear();
medQtyField.clear();
medNameLabel.setText("-");
medPriceLabel.setText("-");
medStockLabel.setText("-");
statusLabel.setText("Item added!");
statusLabel.setStyle("-fx-text-fill: #27ae60;");
} catch (NumberFormatException ex) {
statusLabel.setText("Invalid quantity!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
});
// Totals
totalLabel = new Label("Total: Rs.0.00");
totalLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TextField discountField = new TextField("0");
discountField.setPromptText("Discount %");
discountField.setPrefWidth(80);
netLabel = new Label("Net: Rs.0.00");
netLabel.setFont(Font.font("Arial", FontWeight.BOLD, 16));
netLabel.setStyle("-fx-text-fill: #27ae60;");
HBox totalsBox = new HBox(10);
totalsBox.setAlignment(Pos.CENTER_LEFT);
totalsBox.getChildren().addAll(new Label("Discount %:"), discountField);
// Generate Bill button - uses multithreaded billing
Button generateBtn = new Button("Generate Bill (Threaded)");
```

```
generateBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; " +
"-fx-font-size: 14; -fx-pref-width: 280; -fx-cursor: hand;");
generateBtn.setOnAction(e -> {
if (billItems.isEmpty()) {
statusLabel.setText("Add items first!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
if (custNameField.getText().trim().isEmpty()) {
statusLabel.setText("Enter customer name!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
String billId = IDGenerator.generateBillId();
String date = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
currentBill = new Bill(billId, custIdField.getText().trim(),
custNameField.getText().trim(), date,
userService.getCurrentUser().getUsername());
for (BillItem item : billItems) {
currentBill.addItem(item);
}
try {
double discPct = Double.parseDouble(discountField.getText().trim());
currentBill.applyDiscount(discPct); // Method overloading demo
} catch (NumberFormatException ex) {
// No discount
}
// Multi-threaded billing using BillingThread
statusLabel.setText("Processing bill...");
statusLabel.setStyle("-fx-text-fill: #3498db;");
generateBtn.setDisable(true);
BillingThread billingThread = new BillingThread(billingService, currentBill,
(success, message, bill) -> {
// Use Platform.runLater to update UI from thread
Platform.runLater(() -> {
if (success) {
statusLabel.setText(message + " Invoice generated!");
statusLabel.setStyle("-fx-text-fill: #27ae60;");
billItems.clear();
updateTotals();
custIdField.clear();
custNameField.clear();
} else {
statusLabel.setText(message);
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
}
generateBtn.setDisable(false);
});
});
billingThread.start();
// Demonstrate join() - wait for billing thread in another thread
Thread waitThread = new Thread(() -> {
try {
billingThread.join(); // Wait for billing to complete
System.out.println("Billing thread completed via join()");
} catch (InterruptedException ex) {
```

```
Thread.currentThread().interrupt();
}
});
waitThread.setDaemon(true);
waitThread.start();
});
Button clearBillBtn = new Button("Clear Bill");
clearBillBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor:
hand;");
clearBillBtn.setOnAction(e -> {
billItems.clear();
updateTotals();
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, custGrid,
new Separator(), addItemTitle, itemGrid, addItemBtn, statusLabel,
new Separator(), totalLabel, totalsBox, netLabel,
generateBtn, clearBillBtn);
root.setLeft(formBox);
// Center - Bill items table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
Label tableTitle = new Label("Current Bill Items");
tableTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
billItemsTable = new TableView<>();
billItemsTable.setItems(billItems);
TableColumn<BillItem, String> nameCol = new TableColumn<>("Medicine");
nameCol.setCellValueFactory(new PropertyValueFactory<>("medicineName"));
nameCol.setPrefWidth(200);
TableColumn<BillItem, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
qtyCol.setPrefWidth(60);
TableColumn<BillItem, Double> priceCol = new TableColumn<>("Unit Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("unitPrice"));
priceCol.setPrefWidth(100);
TableColumn<BillItem, Double> subCol = new TableColumn<>("Subtotal");
subCol.setCellValueFactory(new PropertyValueFactory<>("subtotal"));
subCol.setPrefWidth(100);
billItemsTable.getColumns().addAll(nameCol, qtyCol, priceCol, subCol);
// Past bills table
Label pastTitle = new Label("Past Bills");
pastTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
TableView<Bill> pastBillsTable = new TableView<>();
ObservableList<Bill> pastBills =
FXCollections.observableArrayList(billingService.getAllBills());
pastBillsTable.setItems(pastBills);
TableColumn<Bill, String> billIdCol = new TableColumn<>("Bill ID");
billIdCol.setCellValueFactory(new PropertyValueFactory<>("billId"));
TableColumn<Bill, String> billCustCol = new TableColumn<>("Customer");
billCustCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
TableColumn<Bill, String> billDateCol = new TableColumn<>("Date");
billDateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Bill, Double> billAmtCol = new TableColumn<>("Amount");
billAmtCol.setCellValueFactory(new PropertyValueFactory<>("netAmount"));
```

```
pastBillsTable.getColumns().addAll(billIdCol, billCustCol, billDateCol, billAmtCol);
pastBillsTable.setPrefHeight(200);
VBox.setVgrow(billItemsTable, Priority.ALWAYS);
tableBox.getChildren().addAll(tableTitle, billItemsTable, pastTitle, pastBillsTable);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void updateTotals() {
double total = 0;
for (BillItem item : billItems) {
total += item.getSubtotal();
}
totalLabel.setText(String.format("Total: Rs.%.2f", total));
netLabel.setText(String.format("Net: Rs.%.2f", total));
}
public Scene getScene() {
return scene;
}
}
```

26. SupplierScreen.java

Supplier records are managed from this screen. The layout mirrors the customer screen: table on the right showing all suppliers, form on the left for data entry, and a row of four buttons for Add, Update, Delete, and Clear. Clicking a row in the table copies all its fields into the form so the staff member can make changes without retyping everything. A search box above the table narrows results by company name.

Code for SupplierScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Supplier;
import com.pharmacy.service.SupplierService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Supplier Management Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Event handling.
*/
public class SupplierScreen {
private Scene scene;
private PharmacyApp app;
private SupplierService supplierService;
private TableView<Supplier> tableView;
private ObservableList<Supplier> tableData;
public SupplierScreen(PharmacyApp app, SupplierService supplierService) {
this.app = app;
```

```
this.supplierService = supplierService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Supplier Management");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("Add / Edit Supplier");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");
TextField companyField = new TextField();
companyField.setPromptText("Company name");
TextField contactField = new TextField();
contactField.setPromptText("Contact person");
TextField phoneField = new TextField();
phoneField.setPromptText("Phone");
TextField emailField = new TextField();
emailField.setPromptText("Email");
TextField addressField = new TextField();
addressField.setPromptText("Address");
int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Company:"), 0, row); grid.add(companyField, 1, row++);
grid.add(new Label("Contact:"), 0, row); grid.add(contactField, 1, row++);
grid.add(new Label("Phone:"), 0, row); grid.add(phoneField, 1, row++);
grid.add(new Label("Email:"), 0, row); grid.add(emailField, 1, row++);
grid.add(new Label("Address:"), 0, row); grid.add(addressField, 1, row++);
Label statusLabel = new Label();
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
```

```
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
addBtn.setOnAction(e -> {
String id = IDGenerator.generateSupplierId();
Supplier sup = new Supplier(id, companyField.getText(), contactField.getText(),
phoneField.getText(), emailField.getText(), addressField.getText());
supplierService.addSupplier(sup);
refreshTable();
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("Supplier added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
updateBtn.setOnAction(e -> {
String id = idField.getText();
if (id.isEmpty()) { statusLabel.setText("Select a supplier!"); return; }
Supplier sup = new Supplier(id, companyField.getText(), contactField.getText(),
phoneField.getText(), emailField.getText(), addressField.getText());
supplierService.updateSupplier(sup);
refreshTable();
statusLabel.setText("Supplier updated: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
deleteBtn.setOnAction(e -> {
String id = idField.getText();
if (!id.isEmpty()) {
supplierService.deleteSupplier(id);
refreshTable();
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("Supplier deleted.");
}
});
clearBtn.setOnAction(e -> {
clearFields(idField, companyField, contactField, phoneField, emailField, addressField);
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
// Search
HBox searchBox = new HBox(10);
TextField searchField = new TextField();
searchField.setPromptText("Search by company name...");
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String q = searchField.getText().trim();
if (q.isEmpty()) refreshTable();
else {
tableData = FXCollections.observableArrayList(supplierService.searchByName(q));
```

```
tableView.setItems(tableData);
}
});
Button showAllBtn = new Button("Show All");
showAllBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
showAllBtn.setOnAction(e -> refreshTable());
searchBox.getChildren().addAll(searchField, searchBtn, showAllBtn);
tableView = new TableView<>();
TableColumn<Supplier, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("supplierId"));
TableColumn<Supplier, String> compCol = new TableColumn<>("Company");
compCol.setCellValueFactory(new PropertyValueFactory<>("companyName"));
compCol.setPrefWidth(150);
TableColumn<Supplier, String> contCol = new TableColumn<>("Contact");
contCol.setCellValueFactory(new PropertyValueFactory<>("contactPerson"));
contCol.setPrefWidth(120);
TableColumn<Supplier, String> phoneCol = new TableColumn<>("Phone");
phoneCol.setCellValueFactory(new PropertyValueFactory<>("phone"));
phoneCol.setPrefWidth(110);
TableColumn<Supplier, String> emailCol = new TableColumn<>("Email");
emailCol.setCellValueFactory(new PropertyValueFactory<>("email"));
emailCol.setPrefWidth(150);
tableView.getColumns().addAll(idCol, compCol, contCol, phoneCol, emailCol);
refreshTable();
// Table selection listener
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
idField.setText(n.getSupplierId());
companyField.setText(n.getCompanyName());
contactField.setText(n.getContactPerson());
phoneField.setText(n.getPhone());
emailField.setText(n.getEmail());
addressField.setText(n.getAddress());
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(searchBox, tableView);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(supplierService.getAllSuppliers());
tableView.setItems(tableData);
}
private void clearFields(TextField... fields) {
for (TextField f : fields) f.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() { return scene; }
}
```

27. CustomerScreen.java

Patient records live in this screen. Same table-and-form layout as the supplier screen. The search matches against the patient name. When a new customer is added, the ID is generated

automatically using IDGenerator - the staff member does not type it. Pressing Clear empties the form and removes any table row selection so the form is ready for a new entry.

Code for CustomerScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Customer;
import com.pharmacy.service.CustomerService;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
/**
* Customer/Patient Records Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, Event handling.
*/
public class CustomerScreen {
private Scene scene;
private PharmacyApp app;
private CustomerService customerService;
private TableView<Customer> tableView;
private ObservableList<Customer> tableData;
public CustomerScreen(PharmacyApp app, CustomerService customerService) {
this.app = app;
this.customerService = customerService;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Customer / Patient Records");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(320);
```

```
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("Add / Edit Customer");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField idField = new TextField();
idField.setEditable(false);
idField.setPromptText("Auto-generated");
TextField nameField = new TextField();
nameField.setPromptText("Full name");
TextField phoneField = new TextField();
phoneField.setPromptText("Phone number");
TextField emailField = new TextField();
emailField.setPromptText("Email");
TextField addressField = new TextField();
addressField.setPromptText("Address");
int row = 0;
grid.add(new Label("ID:"), 0, row); grid.add(idField, 1, row++);
grid.add(new Label("Name:"), 0, row); grid.add(nameField, 1, row++);
grid.add(new Label("Phone:"), 0, row); grid.add(phoneField, 1, row++);
grid.add(new Label("Email:"), 0, row); grid.add(emailField, 1, row++);
grid.add(new Label("Address:"), 0, row); grid.add(addressField, 1, row++);
Label statusLabel = new Label();
HBox btnBox = new HBox(8);
Button addBtn = new Button("Add");
addBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-cursor: hand;");
Button updateBtn = new Button("Update");
updateBtn.setStyle("-fx-background-color: #2980b9; -fx-text-fill: white; -fx-cursor: hand;");
Button deleteBtn = new Button("Delete");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
Button clearBtn = new Button("Clear");
clearBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
btnBox.getChildren().addAll(addBtn, updateBtn, deleteBtn, clearBtn);
addBtn.setOnAction(e -> {
String id = IDGenerator.generateCustomerId();
Customer cust = new Customer(id, nameField.getText(), phoneField.getText(),
emailField.getText(), addressField.getText());
customerService.addCustomer(cust);
refreshTable();
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("Customer added: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
updateBtn.setOnAction(e -> {
String id = idField.getText();
if (id.isEmpty()) { statusLabel.setText("Select a customer!"); return; }
Customer cust = new Customer(id, nameField.getText(), phoneField.getText(),
emailField.getText(), addressField.getText());
customerService.updateCustomer(cust);
refreshTable();
statusLabel.setText("Customer updated: " + id);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
deleteBtn.setOnAction(e -> {
```

```
String id = idField.getText();
if (!id.isEmpty()) {
customerService.deleteCustomer(id);
refreshTable();
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("Customer deleted.");
}
});
clearBtn.setOnAction(e -> {
clearFields(idField, nameField, phoneField, emailField, addressField);
statusLabel.setText("");
});
formBox.getChildren().addAll(formTitle, grid, btnBox, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
HBox searchBox = new HBox(10);
TextField searchField = new TextField();
searchField.setPromptText("Search by name...");
Button searchBtn = new Button("Search");
searchBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;");
searchBtn.setOnAction(e -> {
String q = searchField.getText().trim();
if (q.isEmpty()) refreshTable();
else {
tableData = FXCollections.observableArrayList(customerService.searchByName(q));
tableView.setItems(tableData);
}
});
Button showAllBtn = new Button("Show All");
showAllBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
showAllBtn.setOnAction(e -> refreshTable());
searchBox.getChildren().addAll(searchField, searchBtn, showAllBtn);
tableView = new TableView<>();
TableColumn<Customer, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("customerId"));
TableColumn<Customer, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Customer, String> phoneCol = new TableColumn<>("Phone");
phoneCol.setCellValueFactory(new PropertyValueFactory<>("phone"));
phoneCol.setPrefWidth(120);
TableColumn<Customer, String> emailCol = new TableColumn<>("Email");
emailCol.setCellValueFactory(new PropertyValueFactory<>("email"));
emailCol.setPrefWidth(170);
TableColumn<Customer, String> addrCol = new TableColumn<>("Address");
addrCol.setCellValueFactory(new PropertyValueFactory<>("address"));
addrCol.setPrefWidth(180);
tableView.getColumns().addAll(idCol, nameCol, phoneCol, emailCol, addrCol);
refreshTable();
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
idField.setText(n.getCustomerId());
nameField.setText(n.getName());
```

```
phoneField.setText(n.getPhone());
emailField.setText(n.getEmail());
addressField.setText(n.getAddress());
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(searchBox, tableView);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(customerService.getAllCustomers());
tableView.setItems(tableData);
}
private void clearFields(TextField... fields) {
for (TextField f : fields) f.clear();
tableView.getSelectionModel().clearSelection();
}
public Scene getScene() { return scene; }
}
```

28. PrescriptionScreen.java

Prescriptions are more complex than other records because they contain nested items. The form on the left starts with the patient details and doctor name. Below that is a mini-form to add one medicine at a time: enter the medicine ID, quantity, and dosage text, then click Add Medicine. Each addition appears in a small list on screen. Once all items are added, clicking Save builds the full prescription object with all its items and sends it to PrescriptionService. Clicking an existing prescription in the main table shows a full text summary in a text area at the bottom, listing every medicine in that prescription.

Code for PrescriptionScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.*;
import com.pharmacy.service.*;
import com.pharmacy.util.IDGenerator;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.Date;
/**
* Prescription Handling Screen.
* Demonstrates: JavaFX TableView, GridPane, VBox/HBox, ArrayList operations.
*/
public class PrescriptionScreen {
private Scene scene;
private PharmacyApp app;
```

```
private PrescriptionService prescriptionService;
private CustomerService customerService;
private InventoryManager inventoryManager;
private TableView<Prescription> tableView;
private ObservableList<Prescription> tableData;
public PrescriptionScreen(PharmacyApp app, PrescriptionService prescriptionService,
CustomerService customerService, InventoryManager inventoryManager) {
this.app = app;
this.prescriptionService = prescriptionService;
this.customerService = customerService;
this.inventoryManager = inventoryManager;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Prescription Handling");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Form
VBox formBox = new VBox(10);
formBox.setPadding(new Insets(15));
formBox.setPrefWidth(330);
formBox.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0 1
0 0;");
Label formTitle = new Label("New Prescription");
formTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
GridPane grid = new GridPane();
grid.setVgap(8);
grid.setHgap(8);
TextField custIdField = new TextField();
custIdField.setPromptText("Customer ID");
TextField custNameField = new TextField();
custNameField.setPromptText("Patient name");
TextField doctorField = new TextField();
doctorField.setPromptText("Doctor name");
TextField notesField = new TextField();
notesField.setPromptText("Notes/Instructions");
custIdField.textProperty().addListener((obs, o, n) -> {
if (n != null && !n.isEmpty()) {
Customer cust = customerService.getCustomerById(n.trim());
if (cust != null) custNameField.setText(cust.getName());
}
});
```

```
int r = 0;
grid.add(new Label("Customer ID:"), 0, r); grid.add(custIdField, 1, r++);
grid.add(new Label("Patient Name:"), 0, r); grid.add(custNameField, 1, r++);
grid.add(new Label("Doctor:"), 0, r); grid.add(doctorField, 1, r++);
grid.add(new Label("Notes:"), 0, r); grid.add(notesField, 1, r++);
// Prescription items
Label itemsTitle = new Label("Prescribed Medicines");
itemsTitle.setFont(Font.font("Arial", FontWeight.BOLD, 13));
TextField medIdField = new TextField();
medIdField.setPromptText("Medicine ID");
Label medNameLabel = new Label("-");
TextField medQtyField = new TextField();
medQtyField.setPromptText("Quantity");
TextField dosageField = new TextField();
dosageField.setPromptText("e.g., 1-0-1");
medIdField.textProperty().addListener((obs, o, n) -> {
if (n != null && !n.isEmpty()) {
Medicine med = inventoryManager.getMedicineById(n.trim());
if (med != null) medNameLabel.setText(med.getName());
}
});
GridPane itemGrid = new GridPane();
itemGrid.setVgap(6);
itemGrid.setHgap(6);
int ir = 0;
itemGrid.add(new Label("Med ID:"), 0, ir); itemGrid.add(medIdField, 1, ir++);
itemGrid.add(new Label("Name:"), 0, ir); itemGrid.add(medNameLabel, 1, ir++);
itemGrid.add(new Label("Qty:"), 0, ir); itemGrid.add(medQtyField, 1, ir++);
itemGrid.add(new Label("Dosage:"), 0, ir); itemGrid.add(dosageField, 1, ir++);
// Items list display
ListView<String> itemsList = new ListView<>();
itemsList.setPrefHeight(120);
java.util.ArrayList<PrescriptionItem> presItems = new java.util.ArrayList<>();
Label statusLabel = new Label();
Button addItemBtn = new Button("Add Medicine");
addItemBtn.setStyle("-fx-background-color: #3498db; -fx-text-fill: white; -fx-cursor: hand;
-fx-pref-width: 200;");
addItemBtn.setOnAction(e -> {
try {
String medId = medIdField.getText().trim();
Medicine med = inventoryManager.getMedicineById(medId);
if (med == null) { statusLabel.setText("Medicine not found!"); return; }
int qty = Integer.parseInt(medQtyField.getText().trim());
PrescriptionItem item = new PrescriptionItem(medId, med.getName(), qty, dosageField.getText());
presItems.add(item);
itemsList.getItems().add(item.toString());
medIdField.clear();
medQtyField.clear();
dosageField.clear();
medNameLabel.setText("-");
} catch (Exception ex) {
statusLabel.setText("Error: " + ex.getMessage());
}
});
Button saveBtn = new Button("Save Prescription");
saveBtn.setStyle("-fx-background-color: #27ae60; -fx-text-fill: white; -fx-font-size: 13; " +
```

```
"-fx-pref-width: 280; -fx-cursor: hand;");
saveBtn.setOnAction(e -> {
if (custNameField.getText().trim().isEmpty() || doctorField.getText().trim().isEmpty()) {
statusLabel.setText("Fill all fields!");
statusLabel.setStyle("-fx-text-fill: #e74c3c;");
return;
}
String presId = IDGenerator.generatePrescriptionId();
String date = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
Prescription pres = new Prescription(presId, custIdField.getText().trim(),
custNameField.getText(), doctorField.getText(), date, notesField.getText());
for (PrescriptionItem item : presItems) {
pres.addItem(item);
}
prescriptionService.addPrescription(pres);
refreshTable();
presItems.clear();
itemsList.getItems().clear();
custIdField.clear(); custNameField.clear(); doctorField.clear(); notesField.clear();
statusLabel.setText("Prescription saved: " + presId);
statusLabel.setStyle("-fx-text-fill: #27ae60;");
});
formBox.getChildren().addAll(formTitle, grid, new Separator(),
itemsTitle, itemGrid, addItemBtn, itemsList, new Separator(),
saveBtn, statusLabel);
root.setLeft(formBox);
// Center - Table
VBox tableBox = new VBox(10);
tableBox.setPadding(new Insets(15));
Label tableTitle = new Label("All Prescriptions");
tableTitle.setFont(Font.font("Arial", FontWeight.BOLD, 15));
tableView = new TableView<>();
TableColumn<Prescription, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("prescriptionId"));
TableColumn<Prescription, String> patCol = new TableColumn<>("Patient");
patCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
patCol.setPrefWidth(130);
TableColumn<Prescription, String> docCol = new TableColumn<>("Doctor");
docCol.setCellValueFactory(new PropertyValueFactory<>("doctorName"));
docCol.setPrefWidth(130);
TableColumn<Prescription, String> dateCol = new TableColumn<>("Date");
dateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Prescription, String> notesCol = new TableColumn<>("Notes");
notesCol.setCellValueFactory(new PropertyValueFactory<>("notes"));
notesCol.setPrefWidth(180);
tableView.getColumns().addAll(idCol, patCol, docCol, dateCol, notesCol);
refreshTable();
// Details area
TextArea detailsArea = new TextArea();
detailsArea.setEditable(false);
detailsArea.setPrefHeight(150);
detailsArea.setPromptText("Select a prescription to view details...");
tableView.getSelectionModel().selectedItemProperty().addListener((obs, o, n) -> {
if (n != null) {
StringBuilder sb = new StringBuilder();
```

```
sb.append("Prescription: ").append(n.getPrescriptionId()).append("\n");
sb.append("Patient: ").append(n.getCustomerName()).append("\n");
sb.append("Doctor: Dr. ").append(n.getDoctorName()).append("\n");
sb.append("Date: ").append(n.getDate()).append("\n");
sb.append("Notes: ").append(n.getNotes()).append("\n\n");
sb.append("Prescribed Medicines:\n");
for (PrescriptionItem item : n.getItems()) {
sb.append(" - ").append(item.toString()).append("\n");
}
detailsArea.setText(sb.toString());
}
});
Button deleteBtn = new Button("Delete Selected");
deleteBtn.setStyle("-fx-background-color: #e74c3c; -fx-text-fill: white; -fx-cursor: hand;");
deleteBtn.setOnAction(e -> {
Prescription sel = tableView.getSelectionModel().getSelectedItem();
if (sel != null) {
prescriptionService.deletePrescription(sel.getPrescriptionId());
refreshTable();
detailsArea.clear();
}
});
VBox.setVgrow(tableView, Priority.ALWAYS);
tableBox.getChildren().addAll(tableTitle, tableView, new Label("Details:"), detailsArea,
deleteBtn);
root.setCenter(tableBox);
scene = new Scene(root, 1100, 700);
}
private void refreshTable() {
tableData = FXCollections.observableArrayList(prescriptionService.getAllPrescriptions());
tableView.setItems(tableData);
}
public Scene getScene() { return scene; }
}
```

29. ReportsScreen.java

The reports section is more of a dashboard than a single report. A menu column on the left has buttons for Sales Report, Stock Report, Expiry Alerts, Low Stock Alerts, Summary, and Backup. The summary view - which is what opens by default - shows colour-coded cards displaying total medicines, total customers, total suppliers, total bills, and total revenue. The expiry and low stock views pull data from the live inventory rather than a cached snapshot. The backup button creates a BackupService instance, starts it in a fresh thread, and shows a spinning progress indicator in the content area until the backup callback fires.

Code for ReportsScreen.java:

```
package com.pharmacy.ui;
import com.pharmacy.model.Bill;
import com.pharmacy.model.Medicine;
import com.pharmacy.service.*;
import javafx.application.Platform;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
```

```
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.control.cell.PropertyValueFactory;
import javafx.scene.layout.*;
import javafx.scene.text.Font;
import javafx.scene.text.FontWeight;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.Date;
/**
* Reports & Alerts Screen - Sales reports, stock alerts, expiry alerts, backups.
* Demonstrates: JavaFX TableView, VBox/HBox, Multithreading integration, Backup service.
*/
public class ReportsScreen {
private Scene scene;
private PharmacyApp app;
private InventoryManager inventoryManager;
private BillingService billingService;
private CustomerService customerService;
private SupplierService supplierService;
private ExpiryAlertThread expiryAlertThread;
private StockAlertThread stockAlertRunnable;
public ReportsScreen(PharmacyApp app, InventoryManager inventoryManager,
BillingService billingService, CustomerService customerService,
SupplierService supplierService,
ExpiryAlertThread expiryAlertThread, StockAlertThread stockAlertRunnable) {
this.app = app;
this.inventoryManager = inventoryManager;
this.billingService = billingService;
this.customerService = customerService;
this.supplierService = supplierService;
this.expiryAlertThread = expiryAlertThread;
this.stockAlertRunnable = stockAlertRunnable;
createScene();
}
@SuppressWarnings("unchecked")
private void createScene() {
BorderPane root = new BorderPane();
root.setStyle("-fx-background-color: #ecf0f1;");
// Top bar
HBox topBar = new HBox(15);
topBar.setPadding(new Insets(12, 20, 12, 20));
topBar.setAlignment(Pos.CENTER_LEFT);
topBar.setStyle("-fx-background-color: #2c3e50;");
Button backBtn = new Button("< Back");
backBtn.setStyle("-fx-background-color: #7f8c8d; -fx-text-fill: white; -fx-cursor: hand;");
backBtn.setOnAction(e -> app.showDashboard(app.getUserService().getCurrentUser()));
Label titleLabel = new Label("Reports & Alerts");
titleLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
titleLabel.setStyle("-fx-text-fill: white;");
topBar.getChildren().addAll(backBtn, titleLabel);
root.setTop(topBar);
// Left - Report categories
VBox leftMenu = new VBox(8);
```

```
leftMenu.setPadding(new Insets(15));
leftMenu.setPrefWidth(200);
leftMenu.setStyle("-fx-background-color: white; -fx-border-color: #bdc3c7; -fx-border-width: 0
1 0 0;");
Label menuTitle = new Label("Report Types");
menuTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
// Content area in center
VBox contentArea = new VBox(10);
contentArea.setPadding(new Insets(15));
Button salesReportBtn = createMenuButton("Sales Report", "#27ae60");
Button stockReportBtn = createMenuButton("Stock Report", "#3498db");
Button expiryAlertBtn = createMenuButton("Expiry Alerts", "#e74c3c");
Button lowStockBtn = createMenuButton("Low Stock Alert", "#e67e22");
Button summaryBtn = createMenuButton("Summary", "#9b59b6");
Button backupBtn = createMenuButton("Backup Data", "#34495e");
// Sales Report
salesReportBtn.setOnAction(e -> showSalesReport(contentArea));
// Stock Report
stockReportBtn.setOnAction(e -> showStockReport(contentArea));
// Expiry Alerts (from alert thread)
expiryAlertBtn.setOnAction(e -> showExpiryAlerts(contentArea));
// Low Stock Alert (from stock thread)
lowStockBtn.setOnAction(e -> showLowStockAlerts(contentArea));
// Summary
summaryBtn.setOnAction(e -> showSummary(contentArea));
// Backup
backupBtn.setOnAction(e -> runBackup(contentArea));
leftMenu.getChildren().addAll(menuTitle, new Separator(),
salesReportBtn, stockReportBtn, expiryAlertBtn,
lowStockBtn, summaryBtn, new Separator(), backupBtn);
root.setLeft(leftMenu);
// Initial content
showSummary(contentArea);
root.setCenter(contentArea);
scene = new Scene(root, 1100, 700);
}
private void showSalesReport(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Sales Report");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
String today = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
Label totalSales = new Label(String.format("Total Sales (All Time): Rs.%.2f",
billingService.getTotalSales()));
totalSales.setFont(Font.font("Arial", FontWeight.BOLD, 14));
totalSales.setStyle("-fx-text-fill: #27ae60;");
Label todaySales = new Label(String.format("Today's Sales: Rs.%.2f",
billingService.getTotalSalesByDate(today)));
todaySales.setFont(Font.font("Arial", FontWeight.BOLD, 14));
Label billCount = new Label("Total Bills: " + billingService.getBillCount());
// Bills table
TableView<Bill> billTable = new TableView<>();
ObservableList<Bill> bills = FXCollections.observableArrayList(billingService.getAllBills());
billTable.setItems(bills);
TableColumn<Bill, String> idCol = new TableColumn<>("Bill ID");
```

```
idCol.setCellValueFactory(new PropertyValueFactory<>("billId"));
TableColumn<Bill, String> custCol = new TableColumn<>("Customer");
custCol.setCellValueFactory(new PropertyValueFactory<>("customerName"));
custCol.setPrefWidth(150);
TableColumn<Bill, String> dateCol = new TableColumn<>("Date");
dateCol.setCellValueFactory(new PropertyValueFactory<>("date"));
TableColumn<Bill, Double> totalCol = new TableColumn<>("Total");
totalCol.setCellValueFactory(new PropertyValueFactory<>("totalAmount"));
TableColumn<Bill, Double> discCol = new TableColumn<>("Discount");
discCol.setCellValueFactory(new PropertyValueFactory<>("discount"));
TableColumn<Bill, Double> netCol = new TableColumn<>("Net Amount");
netCol.setCellValueFactory(new PropertyValueFactory<>("netAmount"));
netCol.setPrefWidth(100);
TableColumn<Bill, String> byCol = new TableColumn<>("Billed By");
byCol.setCellValueFactory(new PropertyValueFactory<>("billedBy"));
billTable.getColumns().addAll(idCol, custCol, dateCol, totalCol, discCol, netCol, byCol);
VBox.setVgrow(billTable, Priority.ALWAYS);
contentArea.getChildren().addAll(title, totalSales, todaySales, billCount, billTable);
}
@SuppressWarnings("unchecked")
private void showStockReport(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Stock Report");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
Label countLabel = new Label("Total Medicines in Inventory: " +
inventoryManager.getMedicineCount());
countLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TableView<Medicine> table = new TableView<>();
ObservableList<Medicine> data =
FXCollections.observableArrayList(inventoryManager.getSortedByName());
table.setItems(data);
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(170);
TableColumn<Medicine, String> catCol = new TableColumn<>("Category");
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Quantity");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
TableColumn<Medicine, String> expCol = new TableColumn<>("Expiry");
expCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expCol.setPrefWidth(100);
table.getColumns().addAll(idCol, nameCol, catCol, qtyCol, priceCol, expCol);
VBox.setVgrow(table, Priority.ALWAYS);
contentArea.getChildren().addAll(title, countLabel, table);
}
@SuppressWarnings("unchecked")
private void showExpiryAlerts(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Expiry Alerts");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
title.setStyle("-fx-text-fill: #e74c3c;");
```

```
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
String currentDate = sdf.format(new Date());
Calendar cal = Calendar.getInstance();
cal.add(Calendar.DAY_OF_MONTH, 30);
String thresholdDate = sdf.format(cal.getTime());
// Expired medicines
ArrayList<Medicine> expired = inventoryManager.getExpiredMedicines(currentDate);
Label expiredTitle = new Label("EXPIRED Medicines (" + expired.size() + ")");
expiredTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
expiredTitle.setStyle("-fx-text-fill: #e74c3c;");
TableView<Medicine> expiredTable = new TableView<>();
expiredTable.setItems(FXCollections.observableArrayList(expired));
addMedicineColumns(expiredTable);
expiredTable.setPrefHeight(200);
// Expiring soon
ArrayList<Medicine> expiringSoon = inventoryManager.getExpiringSoon(currentDate,
thresholdDate);
Label expiringTitle = new Label("Expiring Within 30 Days (" + expiringSoon.size() + ")");
expiringTitle.setFont(Font.font("Arial", FontWeight.BOLD, 14));
expiringTitle.setStyle("-fx-text-fill: #e67e22;");
TableView<Medicine> expiringTable = new TableView<>();
expiringTable.setItems(FXCollections.observableArrayList(expiringSoon));
addMedicineColumns(expiringTable);
VBox.setVgrow(expiringTable, Priority.ALWAYS);
contentArea.getChildren().addAll(title, expiredTitle, expiredTable, expiringTitle,
expiringTable);
}
@SuppressWarnings("unchecked")
private void showLowStockAlerts(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Low Stock Alerts (< 10 units)");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
title.setStyle("-fx-text-fill: #e67e22;");
ArrayList<Medicine> lowStock = inventoryManager.getLowStockMedicines();
Label countLabel = new Label("Low Stock Items: " + lowStock.size());
countLabel.setFont(Font.font("Arial", FontWeight.BOLD, 14));
TableView<Medicine> table = new TableView<>();
table.setItems(FXCollections.observableArrayList(lowStock));
addMedicineColumns(table);
VBox.setVgrow(table, Priority.ALWAYS);
contentArea.getChildren().addAll(title, countLabel, table);
}
private void showSummary(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("System Summary");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
GridPane summaryGrid = new GridPane();
summaryGrid.setVgap(15);
summaryGrid.setHgap(20);
summaryGrid.setPadding(new Insets(20));
addSummaryCard(summaryGrid, 0, 0, "Total Medicines",
String.valueOf(inventoryManager.getMedicineCount()), "#3498db");
addSummaryCard(summaryGrid, 1, 0, "Total Customers",
String.valueOf(customerService.getCustomerCount()), "#9b59b6");
addSummaryCard(summaryGrid, 2, 0, "Total Suppliers",
```

```
String.valueOf(supplierService.getSupplierCount()), "#e67e22");
addSummaryCard(summaryGrid, 0, 1, "Total Bills",
String.valueOf(billingService.getBillCount()), "#27ae60");
addSummaryCard(summaryGrid, 1, 1, "Total Sales",
String.format("Rs.%.2f", billingService.getTotalSales()), "#2c3e50");
String currentDate = new SimpleDateFormat("yyyy-MM-dd").format(new Date());
addSummaryCard(summaryGrid, 2, 1, "Low Stock Items",
String.valueOf(inventoryManager.getLowStockMedicines().size()), "#e74c3c");
// Thread status
Label threadStatus = new Label("Background Threads:");
threadStatus.setFont(Font.font("Arial", FontWeight.BOLD, 14));
Label expiryThreadLabel = new Label("Expiry Alert Thread: " +
(expiryAlertThread != null && expiryAlertThread.isAlive() ? "RUNNING" : "STOPPED"));
expiryThreadLabel.setStyle(expiryAlertThread != null && expiryAlertThread.isAlive() ?
"-fx-text-fill: #27ae60;" : "-fx-text-fill: #e74c3c;");
Label stockThreadLabel = new Label("Stock Alert Thread: RUNNING (daemon)");
stockThreadLabel.setStyle("-fx-text-fill: #27ae60;");
contentArea.getChildren().addAll(title, summaryGrid, new Separator(),
threadStatus, expiryThreadLabel, stockThreadLabel);
}
private void addSummaryCard(GridPane grid, int col, int row, String label, String value, String
color) {
VBox card = new VBox(5);
card.setPadding(new Insets(15));
card.setPrefSize(180, 80);
card.setAlignment(Pos.CENTER);
card.setStyle("-fx-background-color: " + color + "; -fx-background-radius: 8;");
Label valLabel = new Label(value);
valLabel.setFont(Font.font("Arial", FontWeight.BOLD, 20));
valLabel.setStyle("-fx-text-fill: white;");
Label nameLabel = new Label(label);
nameLabel.setStyle("-fx-text-fill: #ecf0f1; -fx-font-size: 12;");
card.getChildren().addAll(valLabel, nameLabel);
grid.add(card, col, row);
}
private void runBackup(VBox contentArea) {
contentArea.getChildren().clear();
Label title = new Label("Data Backup");
title.setFont(Font.font("Arial", FontWeight.BOLD, 18));
Label statusLabel = new Label("Starting backup...");
statusLabel.setStyle("-fx-text-fill: #3498db; -fx-font-size: 14;");
ProgressIndicator progress = new ProgressIndicator();
progress.setPrefSize(50, 50);
contentArea.getChildren().addAll(title, progress, statusLabel);
// Run backup in a separate thread (BackupService implements Runnable)
BackupService backupService = new BackupService(
inventoryManager, customerService, supplierService,
billingService, app.getPrescriptionService());
backupService.setCallback(message -> {
Platform.runLater(() -> {
statusLabel.setText(message);
statusLabel.setStyle("-fx-text-fill: #27ae60; -fx-font-size: 14;");
progress.setProgress(1.0);
Label pathLabel = new Label("Backup files saved in: data/ directory");
```

```
pathLabel.setStyle("-fx-font-size: 12;");
Label filesLabel = new Label(
"Files created:\n" +
" - backup_medicines.txt\n" +
" - backup_customers.txt\n" +
" - backup_suppliers.txt\n" +
" - backup_bills.txt\n" +
" - backup_prescriptions.txt");
filesLabel.setStyle("-fx-font-size: 12;");
contentArea.getChildren().addAll(pathLabel, filesLabel);
});
});
Thread backupThread = new Thread(backupService, "BackupThread");
backupThread.setDaemon(true);
backupThread.start();
}
@SuppressWarnings("unchecked")
private void addMedicineColumns(TableView<Medicine> table) {
TableColumn<Medicine, String> idCol = new TableColumn<>("ID");
idCol.setCellValueFactory(new PropertyValueFactory<>("medicineId"));
TableColumn<Medicine, String> nameCol = new TableColumn<>("Name");
nameCol.setCellValueFactory(new PropertyValueFactory<>("name"));
nameCol.setPrefWidth(150);
TableColumn<Medicine, String> catCol = new TableColumn<>("Category");
catCol.setCellValueFactory(new PropertyValueFactory<>("category"));
TableColumn<Medicine, Integer> qtyCol = new TableColumn<>("Qty");
qtyCol.setCellValueFactory(new PropertyValueFactory<>("quantity"));
TableColumn<Medicine, Double> priceCol = new TableColumn<>("Price");
priceCol.setCellValueFactory(new PropertyValueFactory<>("price"));
TableColumn<Medicine, String> expCol = new TableColumn<>("Expiry");
expCol.setCellValueFactory(new PropertyValueFactory<>("expiryDate"));
expCol.setPrefWidth(100);
table.getColumns().addAll(idCol, nameCol, catCol, qtyCol, priceCol, expCol);
}
private Button createMenuButton(String text, String color) {
Button btn = new Button(text);
btn.setPrefWidth(170);
btn.setStyle("-fx-background-color: " + color + "; -fx-text-fill: white; " +
"-fx-font-size: 12; -fx-cursor: hand; -fx-alignment: center-left;");
return btn;
}
public Scene getScene() { return scene; }
}
```

Util Package (com.pharmacy.util)

30. DataStore.java

This is the generic file storage class used by all five service classes. It is declared as a generic class with a type parameter that must be serialisable, written as DataStore of T where T extends Serializable. This means the same class handles file operations for medicines, customers, suppliers, bills, and prescriptions without any duplication. The saveAll method takes a list and writes it to a binary .dat file using Java's object stream classes. The loadAll method reads it back and returns the list. A separate backupToText method writes the same data as a readable text file using a character-based writer. All methods are locked to prevent a background thread from reading a file

while a save is still in progress.

Code for DataStore.java:

```
package com.pharmacy.util;
import java.io.*;
import java.util.ArrayList;
/**
* Generic data store for serialization and deserialization.
* Demonstrates: Generics, Serialization/Deserialization, File I/O.
*
* @param <T> The type of objects to store (must be Serializable)
*/
public class DataStore<T extends Serializable> {
private String filePath;
public DataStore(String filePath) {
this.filePath = filePath;
}
/**
* Save a list of objects using Serialization (ObjectOutputStream).
* Demonstrates: FileOutputStream, ObjectOutputStream, Serialization.
*/
public synchronized void saveAll(ArrayList<T> items) {
try (FileOutputStream fos = new FileOutputStream(filePath);
ObjectOutputStream oos = new ObjectOutputStream(fos)) {
oos.writeObject(items);
} catch (IOException e) {
System.err.println("Error saving data to " + filePath + ": " + e.getMessage());
}
}
/**
* Load a list of objects using Deserialization (ObjectInputStream).
* Demonstrates: FileInputStream, ObjectInputStream, Deserialization.
*/
@SuppressWarnings("unchecked")
public synchronized ArrayList<T> loadAll() {
File file = new File(filePath);
if (!file.exists()) {
return new ArrayList<>();
}
try (FileInputStream fis = new FileInputStream(filePath);
ObjectInputStream ois = new ObjectInputStream(fis)) {
return (ArrayList<T>) ois.readObject();
} catch (IOException | ClassNotFoundException e) {
System.err.println("Error loading data from " + filePath + ": " + e.getMessage());
return new ArrayList<>();
}
}
/**
* Backup data to a text file using FileWriter.
* Demonstrates: FileWriter usage.
*/
public void backupToText(ArrayList<T> items, String backupPath) {
try (FileWriter writer = new FileWriter(backupPath)) {
writer.write("=== DATA BACKUP ===\n");
writer.write("Source: " + filePath + "\n");
writer.write("Records: " + items.size() + "\n");
```

```
writer.write("====================\n\n");
for (T item : items) {
writer.write(item.toString() + "\n");
}
writer.write("\n=== END OF BACKUP ===\n");
} catch (IOException e) {
System.err.println("Error creating backup: " + e.getMessage());
}
}
/**
* Read backup file content using FileReader.
* Demonstrates: FileReader usage.
*/
public String readBackup(String backupPath) {
StringBuilder content = new StringBuilder();
File file = new File(backupPath);
if (!file.exists()) {
return "No backup found.";
}
try (FileReader reader = new FileReader(backupPath)) {
int ch;
while ((ch = reader.read()) != -1) {
content.append((char) ch);
}
} catch (IOException e) {
System.err.println("Error reading backup: " + e.getMessage());
}
return content.toString();
}
public String getFilePath() {
return filePath;
}
}
```

31. IDGenerator.java

Generates unique IDs for every record type in the system. Five counters, one per record type, start at specific values - medicines at 1000, customers at 2000, suppliers at 3000, bills at 4000, prescriptions at 5000. Each counter increments every time a new ID of that type is requested, giving IDs like MED1001, CUS2001, BIL4001, and so on. Each method is locked so that if two threads request an ID at the same moment, they get different values. An updateCounters method is called at startup so the counters continue from where the saved data left off rather than resetting and producing duplicate IDs.

Code for IDGenerator.java:

```
package com.pharmacy.util;
/**
* Utility class for generating unique IDs.
* Demonstrates: Static methods, synchronized for thread safety.
*/
public class IDGenerator {
private static int medicineCounter = 1000;
private static int customerCounter = 2000;
private static int supplierCounter = 3000;
private static int billCounter = 4000;
private static int prescriptionCounter = 5000;
```

```
// Synchronized to ensure thread-safe ID generation
public static synchronized String generateMedicineId() {
return "MED" + (++medicineCounter);
}
public static synchronized String generateCustomerId() {
return "CUS" + (++customerCounter);
}
public static synchronized String generateSupplierId() {
return "SUP" + (++supplierCounter);
}
public static synchronized String generateBillId() {
return "BIL" + (++billCounter);
}
public static synchronized String generatePrescriptionId() {
return "PRE" + (++prescriptionCounter);
}
// Update counters based on existing data
public static void updateCounters(int medCount, int cusCount, int supCount,
int bilCount, int preCount) {
if (medCount > medicineCounter) medicineCounter = medCount;
if (cusCount > customerCounter) customerCounter = cusCount;
if (supCount > supplierCounter) supplierCounter = supCount;
if (bilCount > billCounter) billCounter = bilCount;
if (preCount > prescriptionCounter) prescriptionCounter = preCount;
}
}
```

32. SampleDataLoader.java

On a fresh installation there is no data, which makes the application look broken. This class solves that by loading a set of demo records on first run: ten medicines covering all eight category types, three customers, and three suppliers. The first thing the method does is check whether medicines already exist. If they do, the method returns immediately without loading anything. This check ensures the demo data loads exactly once and never overwrites real data that was added later.

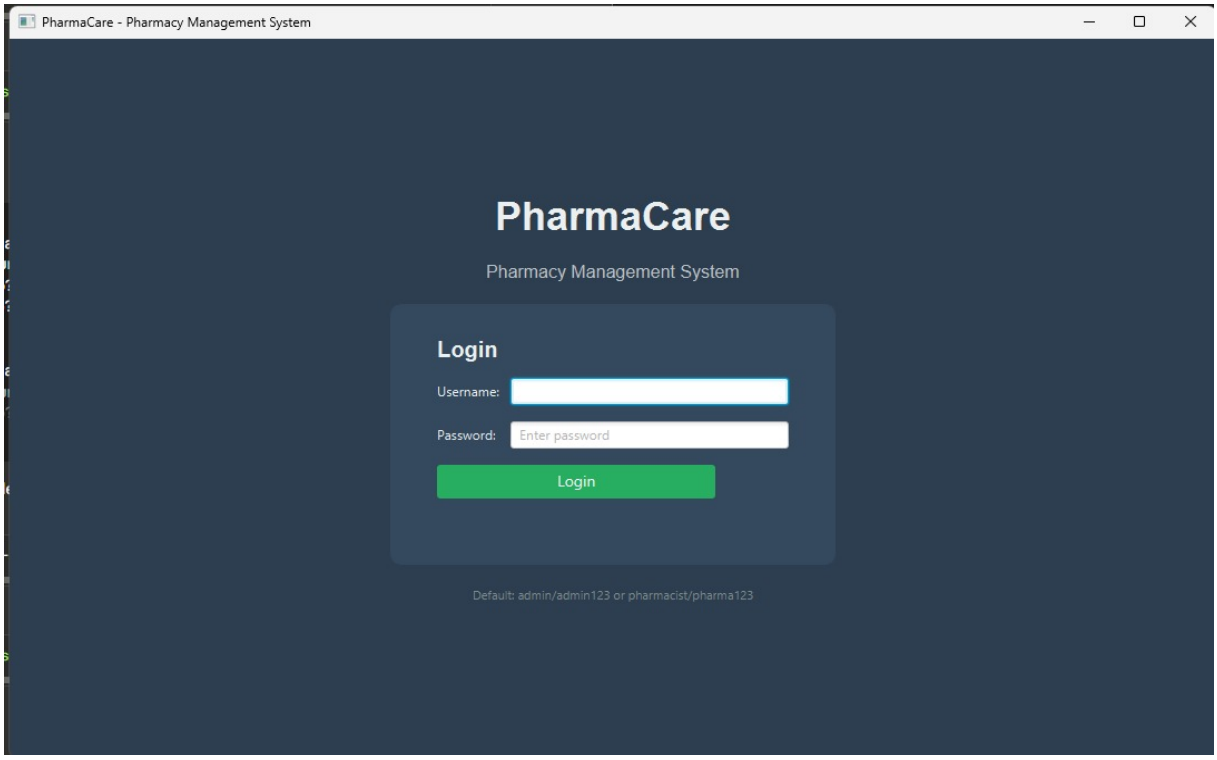
Code for SampleDataLoader.java:

```
package com.pharmacy.util;
import com.pharmacy.model.*;
import com.pharmacy.service.*;
/**
* Loads sample data for demonstration purposes.
* Demonstrates: Object creation, Enum usage, ArrayList operations.
*/
public class SampleDataLoader {
public static void loadSampleData(InventoryManager inventoryManager,
CustomerService customerService,
SupplierService supplierService) {
// Only load if no data exists
if (inventoryManager.getMedicineCount() > 0) {
return;
}
// Sample Suppliers
Supplier s1 = new Supplier("SUP3001", "PharmaCorp India", "Rajesh Kumar",
"9876543210", "rajesh@pharmacorp.com", "Mumbai, Maharashtra");
Supplier s2 = new Supplier("SUP3002", "MediLife Supplies", "Priya Sharma",
```

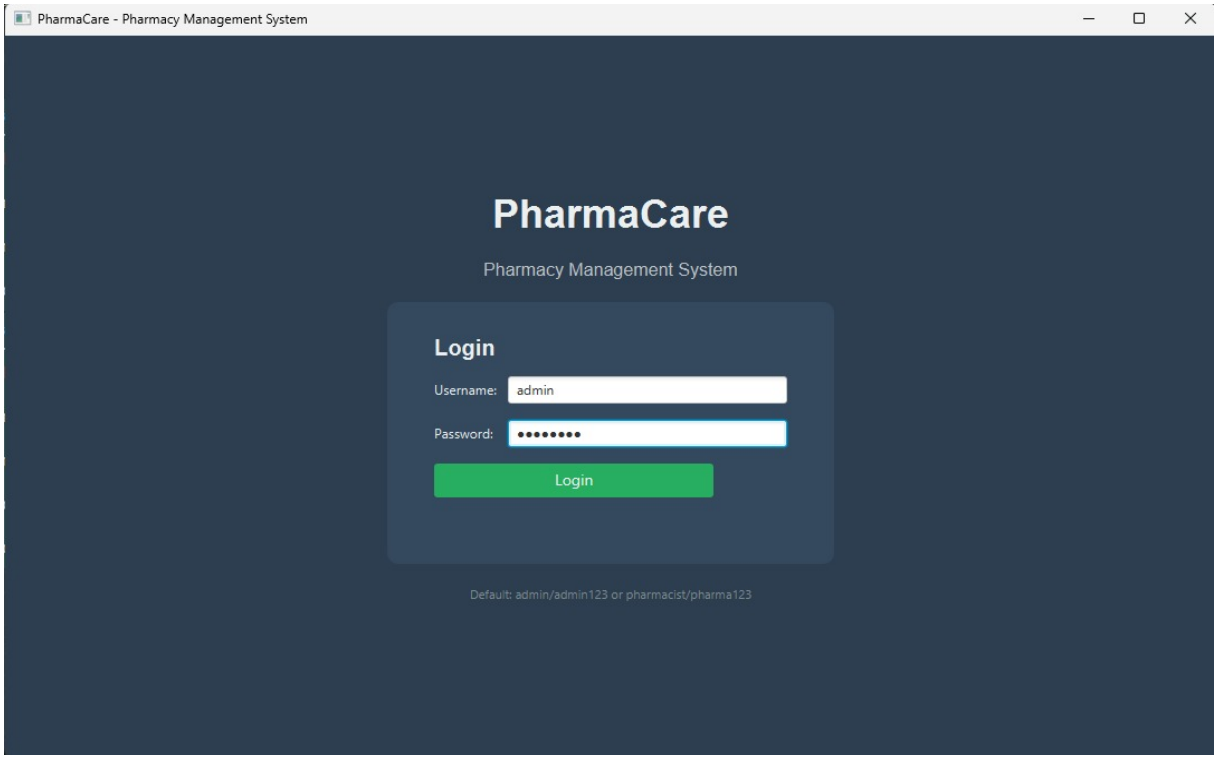
```
"9876543211", "priya@medilife.com", "Delhi, India");
Supplier s3 = new Supplier("SUP3003", "HealthFirst Pharma", "Amit Patel",
"9876543212", "amit@healthfirst.com", "Ahmedabad, Gujarat");

supplierService.addSupplier(s1);
supplierService.addSupplier(s2);
supplierService.addSupplier(s3);

// Sample Medicines
inventoryManager.addMedicine(new Medicine("MED1001", "Paracetamol 500mg", "Sun Pharma",
MedicineCategory.TABLET, "BT001", "2026-12-15", 500, 12.50, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1002", "Amoxicillin 250mg", "Cipla",
MedicineCategory.CAPSULE, "BT002", "2026-06-20", 200, 45.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1003", "Cough Syrup", "Himalaya",
MedicineCategory.SYRUP, "BT003", "2026-03-10", 150, 85.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1004", "Insulin Injection", "Novo Nordisk",
MedicineCategory.INJECTION, "BT004", "2025-08-25", 50, 350.00, "SUP3003"));
inventoryManager.addMedicine(new Medicine("MED1005", "Dettol Ointment", "Reckitt",
MedicineCategory.OINTMENT, "BT005", "2027-01-30", 300, 65.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1006", "Eye Drops", "Alcon",
MedicineCategory.DROPS, "BT006", "2026-09-15", 8, 120.00, "SUP3003"));
inventoryManager.addMedicine(new Medicine("MED1007", "Salbutamol Inhaler", "Cipla",
MedicineCategory.INHALER, "BT007", "2026-11-20", 5, 250.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1008", "ORS Powder", "WHO Standard",
MedicineCategory.POWDER, "BT008", "2027-06-10", 1000, 15.00, "SUP3002"));
inventoryManager.addMedicine(new Medicine("MED1009", "Azithromycin 500mg", "Zydus",
MedicineCategory.TABLET, "BT009", "2026-04-05", 100, 75.00, "SUP3001"));
inventoryManager.addMedicine(new Medicine("MED1010", "Vitamin D3", "HealthVit",
MedicineCategory.CAPSULE, "BT010", "2026-10-30", 400, 30.00, "SUP3003"));

// Sample Customers
customerService.addCustomer(new Customer("CUS2001", "Rahul Verma", "9123456789",
"rahul@email.com", "Bangalore, Karnataka"));
customerService.addCustomer(new Customer("CUS2002", "Sneha Iyer", "9123456790",
"sneha@email.com", "Chennai, Tamil Nadu"));
customerService.addCustomer(new Customer("CUS2003", "Mohammed Ali", "9123456791",
"ali@email.com", "Hyderabad, Telangana"));

// Update ID counters
IDGenerator.updateCounters(1010, 2003, 3003, 4000, 5000);
}
}
```

Results

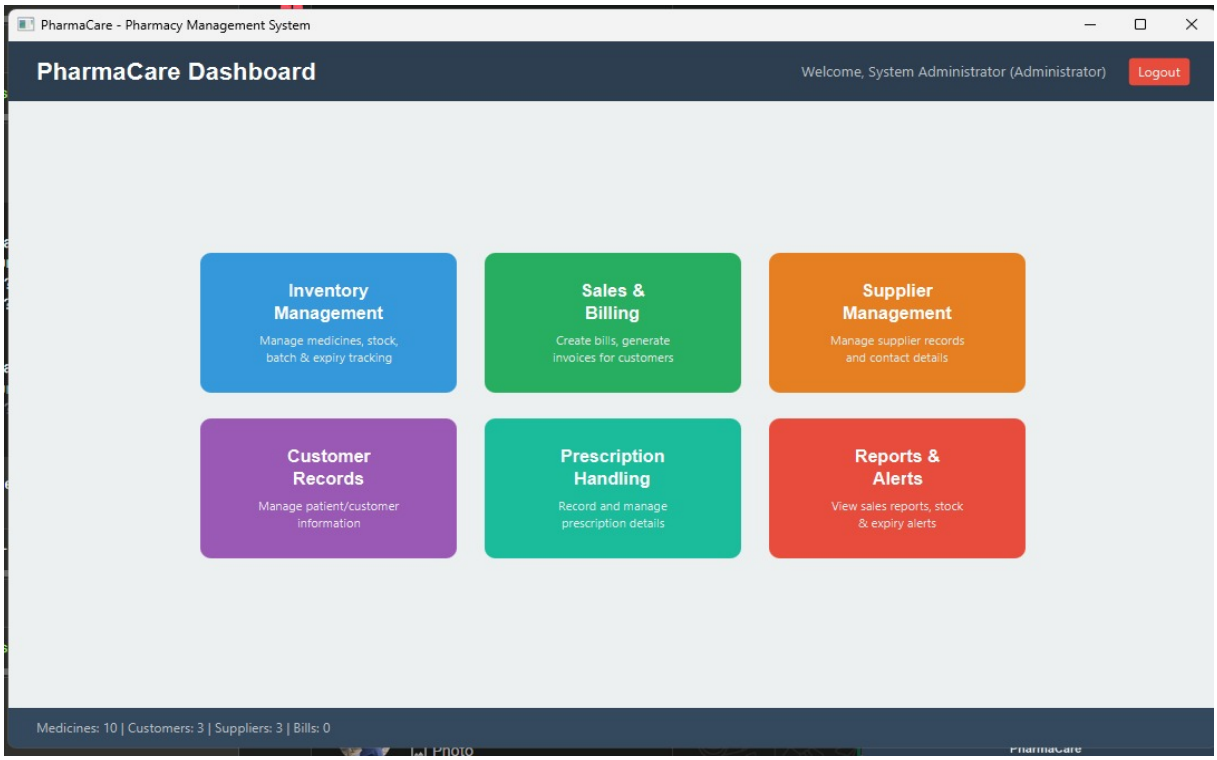
Login Screen



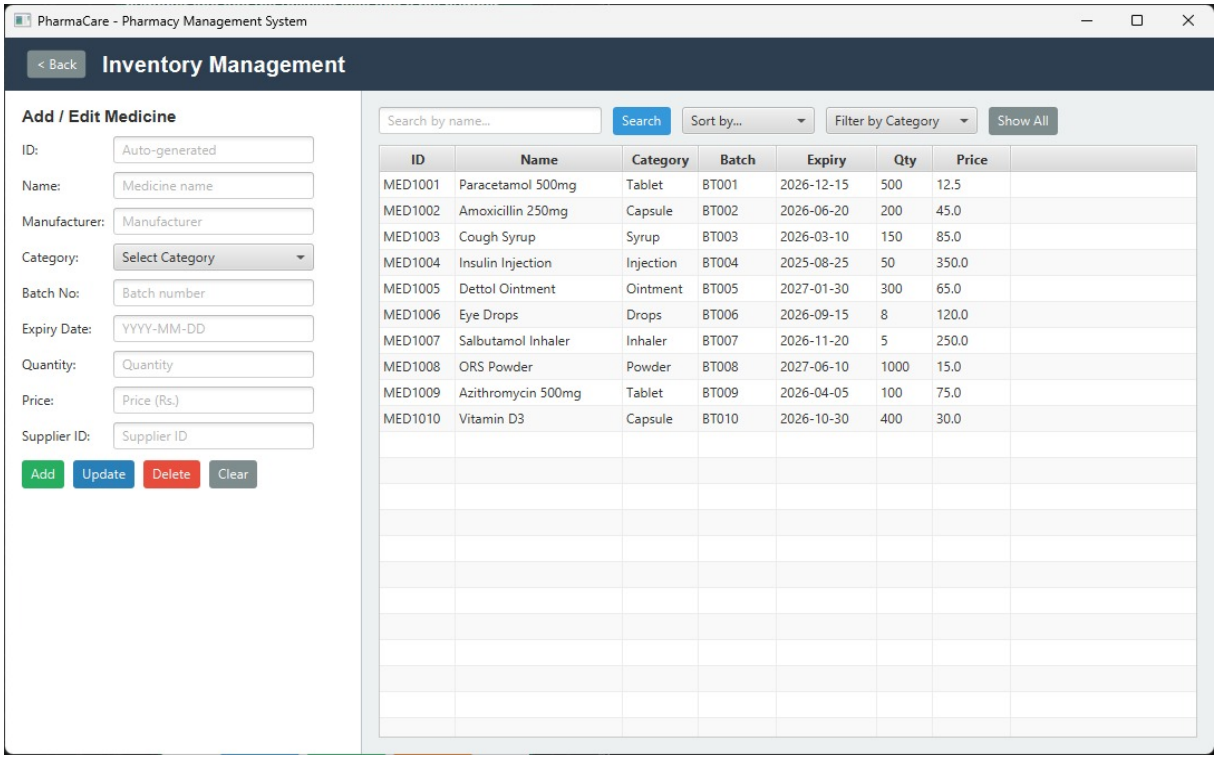
Login Screen - with Credentials



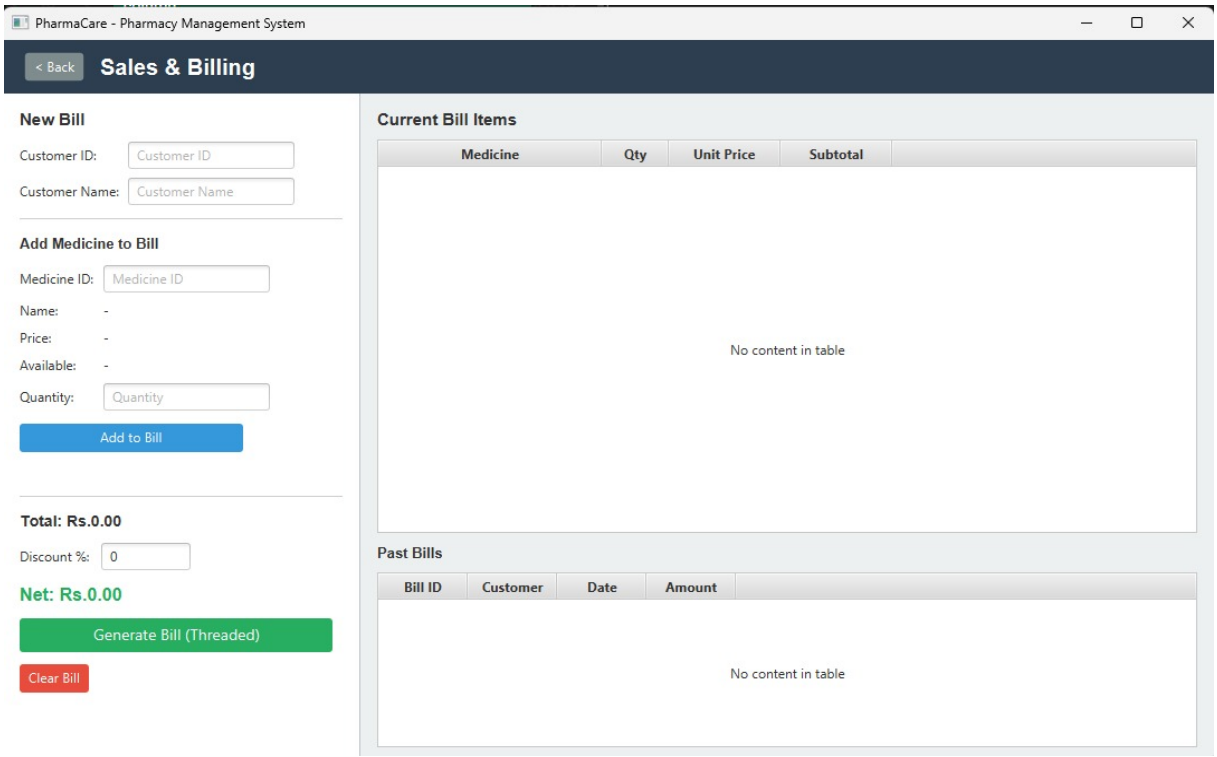
Dashboard Screen



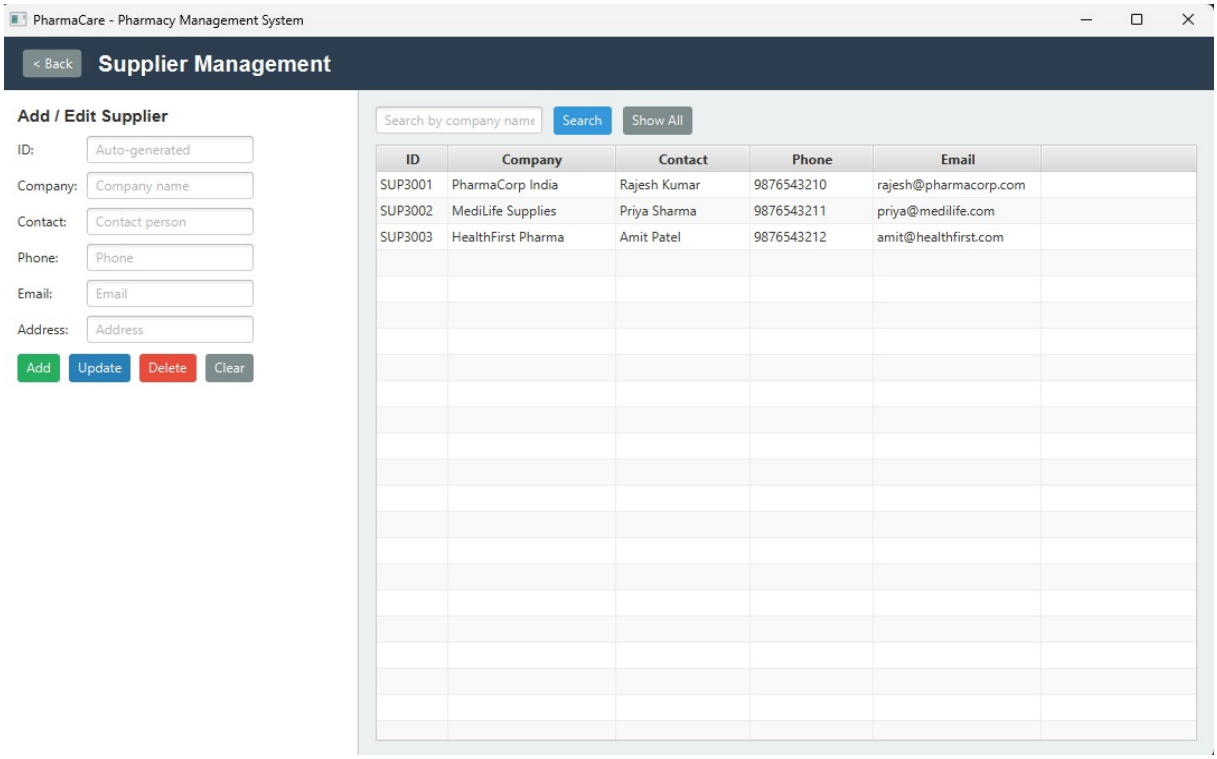
Inventory Management Screen



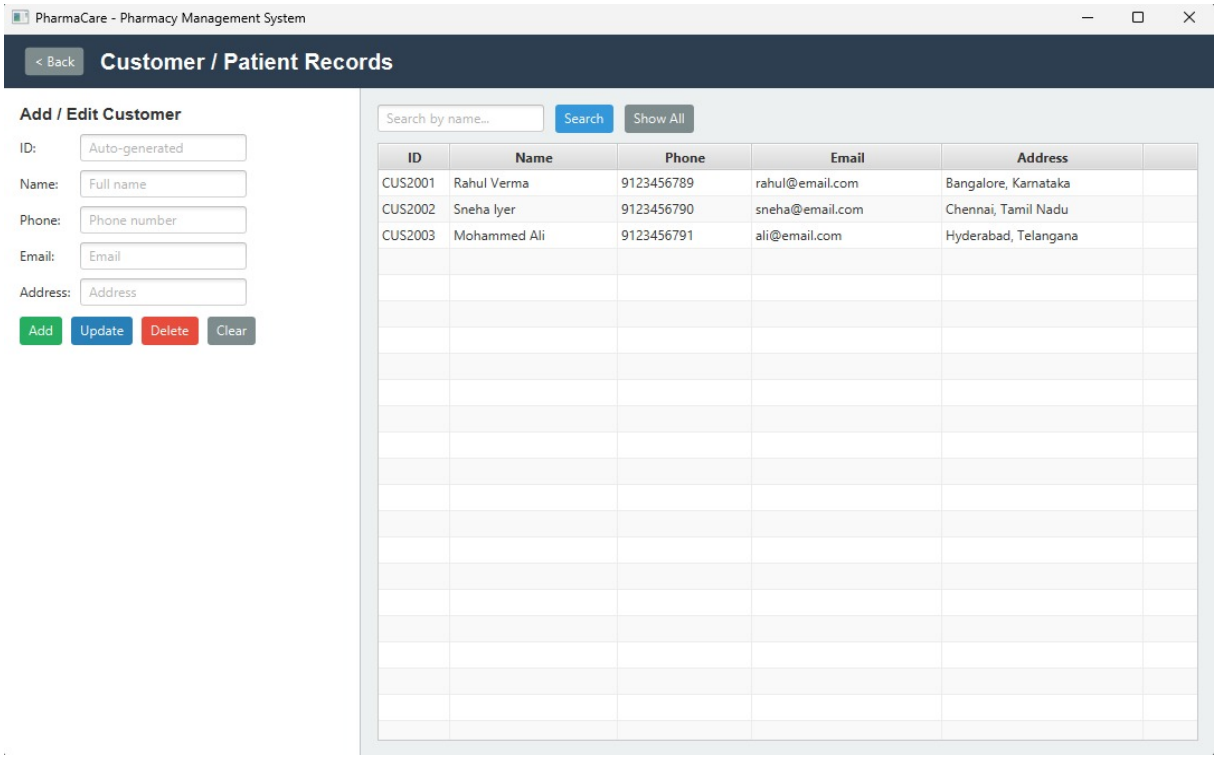
Sales and Billing Screen



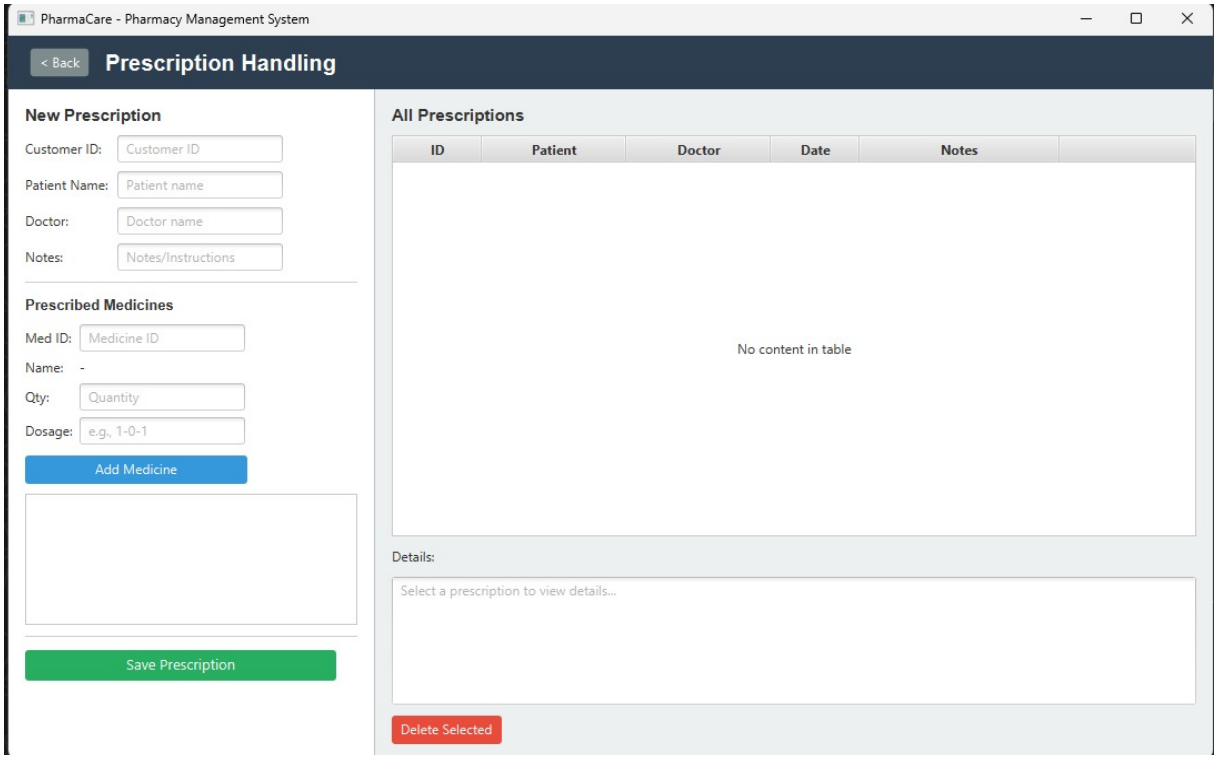
Supplier Management Screen



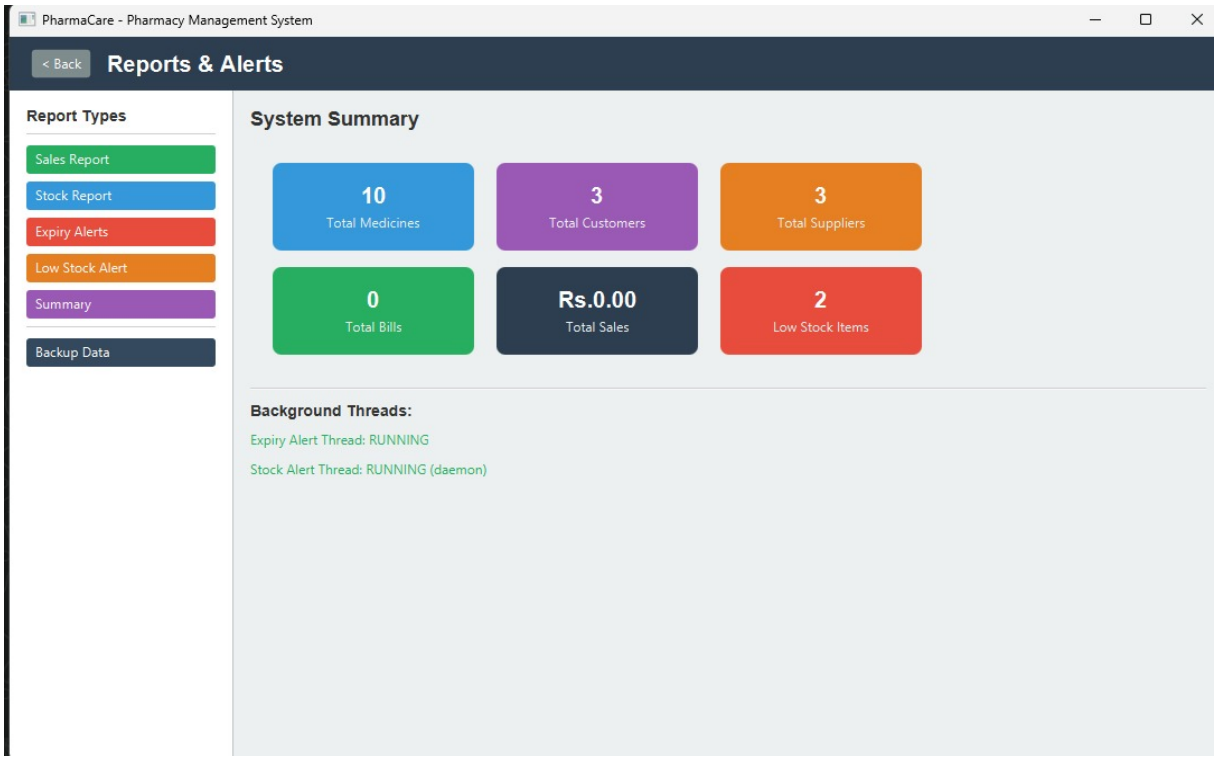
Customer Records Screen



Prescription Handling Screen



Reports and Alerts Screen


PharmaCare - Pharmacy Management System
< Back
Reports & Alerts
Report Types
Sales Report
Stock Report
Expiry Alerts
Low Stock Alert
Summary
Backup Data
System Summary
10
Total Medicines
3
Total Customers
3
Total Suppliers
0
Total Bills
Rs.0.00
Total Sales
2
Low Stock Items
Background Threads:
Expiry Alert Thread: RUNNING
Stock Alert Thread: RUNNING (daemon)

Conclusion

The Pharmacy Management System came together as a working application that we could run, demo with real data, and actually use to process a bill end-to-end. That matters to us because a lot of lab projects stop at "it compiles and the output is correct." This one has a real interface, saves data between sessions, and runs background monitoring without the user doing anything. The design decisions we made early on - separating service logic from screen code, using a generic class for file storage, building the threading around proper daemon threads and volatile flags - all of those paid off when it came to adding the later modules. Adding the prescription screen or the reports section did not require changes to any existing class.

Threading was the part that took the most time to get right. Our first version of the billing screen updated a status label directly from inside the BillingThread. The application crashed sometimes and worked other times - which is exactly the kind of intermittent bug that is hard to track down. After reading about JavaFX's threading rules, we understood that UI components can only be modified from the main application thread. The fix - wrapping the label update inside a call that schedules it to run on the correct thread - was a one-line change that made the crashes disappear entirely. It was a frustrating bug to hit but a genuinely useful thing to understand.

Looking at the complete file list, every OOP concept from the course syllabus is present and used for a real reason. Encapsulation in the model classes keeps fields from being set to invalid values from outside. Inheritance appears in the Thread subclasses and in extending Application. Polymorphism shows up in the overloaded discount methods and in the callback interfaces shared between threads and UI classes. Generics in DataStore removed four copies of identical persistence code. Enums in MedicineCategory and UserRole removed whole categories of runtime bugs. The project ended up being a solid demonstration of why these concepts exist, not just what they are called.

Individual Contributions

Kenneth Martin, 240962047, Roll No. 8

- PharmacyApp.java
- LoginScreen.java
- DashboardScreen.java
- UserService.java
- User.java / UserRole.java

Sam Thomas, 240962242, Roll No. 18

- Medicine.java / MedicineCategory.java
- InventoryManager.java
- InventoryScreen.java
- SampleDataLoader.java

Mohammed Rayyan Sheik, 240962073, Roll No. 12

- BillingService.java
- BillingThread.java
- BillingScreen.java
- Bill.java / BillItem.java

Vedank Singh, 240962043, Roll No. 7

- DataStore.java
- IDGenerator.java
- CustomerService.java / SupplierService.java
- PrescriptionService.java
- ExpiryAlertThread.java / StockAlertThread.java
- BackupService.java
- ReportsScreen.java
- Supplier.java / Customer.java / Prescription.java / PrescriptionItem.java / BillItem.java

We split the work by module but spent a lot of time in each other's code anyway. When Kenneth was building the login screen he needed the User class and UserRole enum that Sam was still writing - so they sat together and finalised those first before the login screen could be completed. Rayyan could not test the billing screen until Vedank had the DataStore class working because nothing could be saved otherwise. That kind of dependency forced us to communicate constantly, which turned out to be good for the project. We caught mismatches in method signatures early, agreed on naming conventions together, and reviewed each other's code before merging. The result is a codebase that is consistent throughout - not four separate pieces written by four people independently, but one system built by a team.

References

- 1. JavaFX Official Documentation: https://openjfx.io/
- 2. JavaFX Tutorials and Codes: https://www.geeksforgeeks.org/tag/javafx/
- 3. Java Serialization Guide: https://www.baeldung.com/java-serialization
- 4. Java Collections Framework: https://docs.oracle.com/javase/8/docs/technotes/guides/collections/overview.html
- 5. Java Multithreading: https://www.geeksforgeeks.org/multithreading-in-java/